Economic insecurity, greenlash and the demand for climate policies in Europe*

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December 8, 2024

Abstract

We study how perceived economic insecurity affects people's support for Pigouvian climate policies across European regions. By developing a new political economy model and using a new regional dataset with 17 European countries, we show that larger perceived economic insecurity is robustly associated with lower support for climate policies. This result occurs because Pigouvian taxes reduce people's purchasing power by increasing the price of private goods; at the same time, feelings of economic insecurity exacerbate voters' concerns about their future income, propelling them to oppose climate policies to protect their economic stability. We also find that voters with 'pro-environmental' attitudes express stronger support for climate policies than those with 'anti-environmental' attitudes; however, 'pro-environmental' voters experience a stronger backlash against climate policies (or "greenlash") than 'anti-environmental' voters when they feel economically insecure. Indeed, 'pro-environmental' voters have a stronger incentive to make a face-about and prioritise their economic concerns over their ideological stance if they feel economically insecure. Finally, perceived economic insecurity affects support for climate policies only for respondents living in European regions with high long-term unemployment. This finding suggests that voters living in regions with structurally weaker labour markets may feel more exposed to economic insecurity.

Keywords: Economic insecurity, Climate change, Climate policies, Public support, Greenlash, Environmental taxes

JEL Codes: Q3, R11, H23, D72

*Thanks to Nawid Siassi for early discussion about this paper. We also thank Christopher Jepsen and other participants of the ESRI lunch seminar for their helpful comments. Errors are our own.

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1 Introduction

Climate change represents the most critical global problem of the 21st century. In response to the climate crisis, the Paris Agreement, established during the United Nations Framework Convention on Climate Change (2015), saw the commitment of more than 190 Parties worldwide to enact public policies focused on limiting the global average temperature growth.[1] More recently, the European Commission approved the European Green Deal to reach the zero-emission target by the year 2050 in the EU countries.[2] These international agreements lean on scientific evidence highlighting the need for greater policy effort to reduce global emissions and the expected rise in temperature (e.g., Höhne et al. (2021)). Yet, despite the scientific community's call and the international agreements for emissions reduction policies, governments continue to face challenges in implementing these policies due to the lack of voters' support (e.g., Drews and Van den Bergh (2016)).[3]

The literature on attitudes towards climate policies is extensive, encompassing different predictors of preferences for climate policies.[4] Among these predictors, a positive cost-benefit perception of climate policies represents one of the most important factors explaining higher support for climate policies. For instance, a recent survey involving respondents from the 20 most CO_2-polluting countries highlights that support for climate policies mostly depends on

[1] Precisely, the agreement focused on reducing "[...] the increase in the global average temperature to well below 2°C above pre-industrial levels" and striving "[...] to limit the temperature increase to 1.5°C above pre-industrial levels" (United Nations Framework Convention on Climate Change, 2015, Article 2, 1 (a)).

[2] For more info, read: `https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/story-von-der-leyen-commission/european-green-deal_en`.

[3] For instance, recently the European Climate Neutrality Observatory (2024) has presented a first report on how European countries are dealing with the zero-emission target. The authors highlight that many countries and sectors are not "on track" with the Green Deal's targets.

[4] Some predictors are, for instance, ideology and environmental concern (e.g., Davidovic et al. (2020); Anderson et al. (2023)), perceived government inefficiency and political mistrust (e.g., Hammar and Jagers (2006); Harring and Jagers (2013); Rafaty (2018); Umit and Schaffer (2020); Davidovic et al. (2020)), intolerance towards other people (Johansson et al., 2022), perceived equity and effectiveness (e.g., Maestre-Andrés et al. (2019); Sommer et al. (2022); Dechezleprêtre et al. (2022)), awareness of the amount of pollution emitted by different income levels (Beiser-McGrath and Busemeyer, 2023) and people's beliefs and "tax aversion" (e.g., Kallbekken et al. (2011); Douenne and Fabre (2022); Andreassen et al. (2024)). See Drews and Van den Bergh (2016), Köppl and Schratzenstaller (2023) and Baiardi (2023) for a more comprehensive review and Bergquist et al. (2022) for a meta-analysis of the determinants of attitude towards climate change and climate policies.

the perceived distributional equity and effectiveness of these policies (Dechezleprêtre et al., 2022). However, although a growing literature highlights the effect of economic insecurity on people's political attitudes (e.g., Algan et al. (2017); Scheiring et al. (2024); Guiso et al. (2024)), the role of income uncertainty and economic insecurity on attitudes towards climate policies has not received much attention in theoretical and empirical research in economics.

In this paper, we present a voting model highlighting the role of economic insecurity on people's preferences for climate policies.[5] In the model, agents vote on a Pigouvian tax on pollution to finance a 'green public' good. Voters exhibit heterogeneous environmental ideologies and face uncertainty about their economic conditions, which represents a source of economic insecurity. When economic insecurity increases — meaning that voters are more likely to become low-income —, agents show less support for environmental taxes, as they are concerned about the erosion of their purchasing power. Notably, a Pigouvian tax would increase the price of private goods, and economic insecurity would lead agents to expect a lower income; both factors reduce agents' expected purchasing power and increase their opposition to environmental taxes. Furthermore, we also show that when economic insecurity is high, opposition to climate policies is weaker among voters with strong 'anti-environmental' ideology. This result arises because 'anti-environmental' voters base their opposition mainly on ideological views rather than economic reasons; therefore when economic insecurity arises, they have a lower marginal utility in shifting their preferences towards a more intense opposition to climate policies. On the other hand, 'pro-environmental' voters — who were initially more supportive of environmental policies— experience a stronger "greenlash" as they gain a higher marginal value from shifting their voting preferences from an 'ideological' to a more 'bread-and-butter' political choice.[6]

We conclude our theoretical model by studying the welfare implications of politically

[5]For simplicity, throughout this paper, we will use "climate policies", "environmental policies", "green policies", "environmental taxes" and "taxes on pollution" interchangeably.

[6]We borrow the term "greenlash" from Tocci (2023). The term recently gained prominence (e.g., see Chapron (2024)) after Tocci (2023) used it to describe a growing opposition to green policies in Europe. However, a similar phenomenon was already observed globally in the 1990s and was similarly termed "green backlash" by Rowell (1996).

supported climate policies. In our welfare analysis, we show that the welfare effect of the voting equilibrium depends on the skewness of the ideology distribution. If the ideology distribution is skewed towards 'anti-environmental' ideology, voters will underfinance the social-optimal level of environmental policies. Conversely, if the ideology distribution is skewed towards 'pro-environmental' ideology, voters will overfinance environmental policies. In both cases, a democratic voting equilibrium can lead to an inefficient allocation of resources to environmental policies and welfare losses due to voters' ideological views.

In the second part of the paper, we bring our model to the data and provide descriptive empirical evidence from European countries. To do so, we build a new dataset providing individual socioeconomic information and aggregate regional economic variables for 17 European countries. Our empirical analysis shows that perceived economic insecurity decreases the support for environmental policies. In line with the model's predictions, the heterogeneous analysis shows that this effect is driven by voters with 'pro-environmental' attitudes. We also find that the respondents' concern about higher costs for energy goods — and, therefore, the erosion of respondent purchasing power— is a potential channel through which economic insecurity affects preferences for climate policies. Finally, we find that individual perceived economic insecurity depends on the surrounding economic environment. In fact, perceived economic insecurity significantly affects preferences for climate policies only in European regions with high long-term unemployment rates. This finding supports the "it's the economy, stupid" stance, namely, a view for which certain policies — including climate policies — are more difficult to implement if economic conditions are not favourable to the voters.[7]

The contribution of our paper is three-fold. First, we contribute to the theoretical literature studying political preferences for climate policies. For instance, Cremer et al. (2004) build a voting model highlighting the importance of refunding rules in shaping voters' preferences for environmental taxes.[8] Karakas and Mitra (2020) theoretically highlight the role

[7]The expression "it's the economy, stupid" was coined by James Carville in 1992, when he was the strategist of Bill Clinton in the 1992 U.S. presidential election. The expression captured the importance of the economy in voter's electoral preferences and decisions.

[8]For other models focusing on the effect of refunding rules on preferences for climate policies, see, for

of ideology by showing that voters' ideological polarisation leads to a divergence in the objectives of climate policies presented by progressive and conservative candidates. Habla and Roeder (2013) develop and calibrate a voting model based on an overlapping generation (OLG) model to study the intergenerational effect of green reforms implemented in Germany in the late 1990s and early 2000s.[9] More recently, Young-Brun (2023) calibrates a voting model to show the impact of urban-rural economic disparities on the political support for a carbon tax in Europe. With our model, we complement this literature by showing that economic insecurity reduces voting support for environmental taxes. We also show that voters with a stronger 'anti-environmental' ideology show a more moderate opposition to environmental taxes following an increase in economic insecurity. We argue that this result occurs because more 'anti-environmental' voters oppose environmental taxes mostly for ideological rather than economic reasons. When economic insecurity increases, they gain a lower marginal value from shifting their preferences towards an even lower environmental tax and higher emissions. Instead, 'pro-environmental' voters experience a stronger "greenlash" reaction. In fact, when 'pro-environmental' voters feel economically insecure, they evaluate their economic motive as marginally more important than their ideological beliefs and move their preferences toward a more 'bread-and-butter' motivated preferred environmental tax.

A second contribution pertains to the empirical literature on economic insecurity and environmental preferences. Most of this literature relies on aggregate variables — such as regional or national unemployment rates— as proxies of economic insecurity, finding a negative effect on preferences for climate policies (e.g., Scruggs and Benegal (2012); Kenny (2020); Arndt et al. (2023)).[10] Yet, little is known about the individual perception of economic insecurity and its interaction with the surrounding economic environment. To date, the only empirical work explicitly addressing the problem of perceived insecurity is Kachi et al. (2015). The authors show that people's expectations about their future income do not significantly affect

instance, Cremer et al. (2007, 2008),Fredriksson and Sterner (2005) and Aidt (2010).

[9]See also Habla and Roeder (2017) and Rausch and Yonezawa (2023) for other examples of majority voting OLG models studying the political support for environmental policies.

[10]We found only one study reporting no effect (i.e., Mildenberger and Leiserowitz (2017)).

preferences for climate policies in the U.S. and Germany.[11] Nonetheless, Kachi et al. (2015)'s results are limited to only two countries and do not show the effect of the interaction between the surrounding economic environment and respondents' economic insecurity on preferences for climate policies.[12] Moreover, in a recent paper, Bossert et al. (2023) argue that measures of perceived economic insecurity should also capture people's experience of past income changes and not only future concerns. With our paper, we extend the previous literature by (i) using a proxy for perceived economic insecurity incorporating past income changes and (ii) expanding the analysis to 17 European countries. Differently from Kachi et al. (2015), we find that higher perceived economic insecurity robustly predicts low political support for climate policies; however, these results are driven by (i) respondents residing in European regions with high long-term unemployment, (ii) respondents with strong 'pro-environmental' attitudes, as predicted by our model.

Finally, our paper brings new empirical evidence of the economic motives behind preferences for environmental policies. The empirical literature studying preferences for environmental policies shows that the correlation with individual income is positive but small or weakly statistically significant (e.g., Levi (2021); Bergquist et al. (2022)), if not statistically insignificant (e.g., Harring and Jagers (2013), Rhodes et al. (2017)). Unlike income, we show that higher perceived economic insecurity is a robust predictor of lower support for climate policies. Our empirical analysis presents a net distinction between the effect of income levels and economic insecurity on preferences for climate policies. As previously written by Bossert et al. (2023), "[economic] insecurity does not refer to the levels of resources but rather to their changes; an individual may be rich, and belong to the dominant cultural population group, but still be [economically] insecure due to their experience of income changes" (p. 804). With our empirical analysis, we show that economically insecure people are not people

[11] The study also shows that optimistic expectations about macroeconomic trends are statistically significantly correlated with a larger "willingness to pay" for climate policies in the U.S. but not in Germany.

[12] The authors also include state-level unemployment rates in a robustness check and find statistically insignificant results. However, they do not study the interaction between perceived economic insecurity and state-level unemployment.

with lower incomes; rather, they are those incorporating their past income changes into their future economic expectations. If people are concerned about experiencing similar economic instability as in the past, they will feel financially insecure — independently of their income — and oppose climate policies to avoid the personal monetary costs of implementing such policies.

The remainder of the paper is structured as follows: in Section 2 we present — using different data sources— some stylised facts across Europe showing a negative cross-country correlation between perceived economic insecurity and support for green policies. In Sections 3–5, we present a simple theoretical political-economy model and a welfare analysis underlining the mechanisms through which economic insecurity affects preferences for climate policies. In Sections 6–7, we bring the model to the data. Finally, in Section 8, we conclude presenting some policy discussion and the possible direction of future research.

2 Stylised facts across Europe

The last 20 years in Europe have been characterised by high economic insecurity (e.g., Ranci et al. (2021) and OECD (2023a)). While economic upheavals became more frequent among European countries, the living standards did not bounce back as rapidly, with only one-fifth of households experiencing stable income growth (OECD, 2023b). During this period, economic instability became widespread along all the income distribution (OECD, 2023b), eventually affecting a wide range of people's political preferences (e.g., Algan et al. (2017); Scheiring et al. (2024); Guiso et al. (2024)). Henceforth, we present some stylised facts showing that different proxies of individual economic insecurity measured in the last 20 years are associated with lower support for environmental policies across European countries.

We begin our descriptive analysis by showing in Figure 1 the cross-country correlation between the average concern about job loss and support for higher energy prices and taxes in 2005-2020. Over this period, Europe has gone through different shocks that have increased

economic insecurity, such as the global financial crisis (2007-2008), the European debt crisis (2009-2010), the European migration crisis (2015) and the most recent Covid-19 pandemic and surge in inflation.[13] As shown in Figure 1, countries with higher average insecurity about their job stability are less likely to support environmental policies involving higher costs, such as price increases and higher environmental taxes.[14]

People's concern about their job instability is not the only proxy for economic insecurity. Using data from the Eurobarometer wave on "fairness perception of the green transition" (European Commission, 2023), we adopt a different measure of economic insecurity, namely the difficulty of paying bills over the last 12 months.[15] In Figure 2, we show the cross-country correlation between the percentage of people facing difficulties in paying their bills and the percentage of people willing to pay higher energy prices to favour a green transition.[16] The regression shows that countries with a higher percentage of people reporting economic insecurity also have a smaller percentage of people supporting Europe's green transition.

This wave of Eurobarometer (European Commission, 2023) also allows to understand why

[13]We include the European migration crisis because people often associate immigration with job insecurity. Immigration has been shown to affect voters' behaviour as it is related to higher perceived labour-market insecurity or concerns about the economic sustainability of the welfare state (e.g., Halla et al. (2017); Margalit (2019)). The concern for immigration-induced labour-market insecurity can also affect political institutions, even if immigration stems from programmes sponsored by the government itself. For instance, the *Bracero Program* was a government programme started in the 1940s to recruit Mexican immigrants to work in the U.S. agricultural sector. The programme reached an average of about 200.000 incoming Mexican workers per year (Ngai, 2014, p. 139) and "[...] was discontinued in 1964, when [The United States] Congress became *concerned about immigrants taking American jobs* [emphasis added]" (Johnson and Acemoglu, 2023, p. 251).

[14]Data from ISSP Environmental datasets for years 2010 (ISSP Research Group, 2019) and 2020 (ISSP Research Group, 2023) and Work Orientation datasets for years 2005 (ISSP Research Group, 2013) and 2015 (ISSP Research Group, 2017). The feeling of job insecurity is measured using the question from the ISSP Work Orientation dataset: "*To what extent, if at all, do you worry about the possibility of losing your job* with reversed answers going from 1 ("*I do not worry at all*") to 4 ("*I worried a great deal*"). The questions from the ISSP Environmental datasets follow the structure: "*And how willing would you be to pay much higher taxes [prices] in order to protect the environment?*", with answers going from 1 ("*Very willing*") to 4 ("*Very unwilling*").

[15]The question is: "*During the last twelve months, would you say you had difficulties to pay your bills at the end of the month?*". We assigned a value of 1 if the respondent answers "*Most of the time*" or "*From time to time*", and 0 if the answer is "*Almost never/ Never*".

[16]The question is "Would you be willing to pay higher energy prices if that helps to speed up the green transition?". The answers are (i) "*Yes, up to 10% more*", (ii) "*Yes, up to 20% more*", (iii) "*Yes, up to 30% more*", (iv) "*Yes, above 30% more*", (v) "*No, you are not willing to pay more as you cannot afford to do so*" and (vi) "*No, you are not willing to pay more even though you could afford to do so*". We re-coded the answer, assigning a value of 1 if the respondent answered "*Yes, [...]*" and 0 if the respondent answered "*No, [...]*".

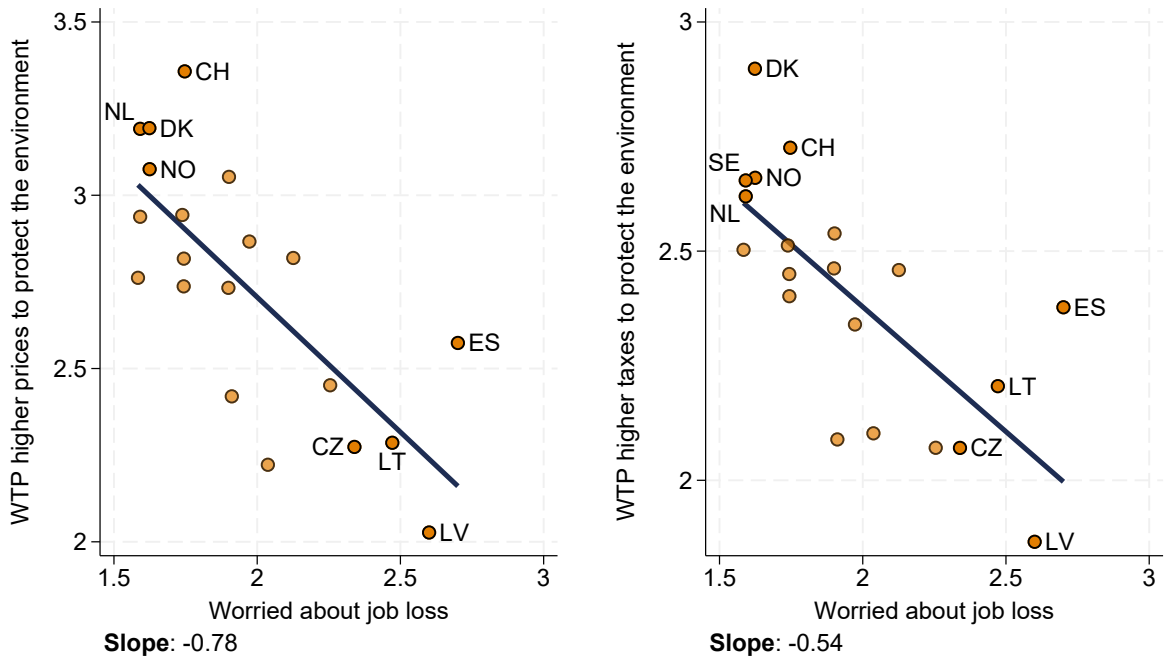

Figure 1: Support for environmental policy and job insecurity among European countries

Notes: the figure shows the cross-country correlation between the average concern about job loss and support for higher energy prices (p-value<0.01) (left) and taxes (p-value<0.01) (right) in Europe, over the period 2005-2020. The Likert scale for job insecurity goes from 1 ("*I do not worry at all*") to 4 ("*I worried a great deal*"). The Likert scale for willingness to pay (WTP) to protect the environment is from 1 ("*Very willing*") to 4 ("*Very unwilling*"). Countries: Austria, Belgium, Switzerland, Czech Republic, Germany, Denmark, Spain, Finland, France, United Kingdom, Hungary, Iceland, Lithuania, Latvia, Netherlands, Norway, Portugal, Sweden, Slovenia, Slovakia.

people do not want to pay for a green transition. In the survey, respondents unwilling to pay higher energy prices to support a European green transition were given two answers: (i) "*No, you are not willing to pay more as you cannot afford to do so*" or (ii) "*No, you are not willing to pay more even though you could afford to do so*". In Figure 2, we aggregated these answers to show the average opposition to higher prices to finance a green transition. Instead, in Figure 3, we show that when people do not want to pay higher energy prices because they cannot afford it, the correlation between economic insecurity is negative and strongly statistically significant (i.e., orange continuous regression line). Nonetheless, a portion of the population opposes the green transition for other non-economic reasons (i.e., the blue dashed

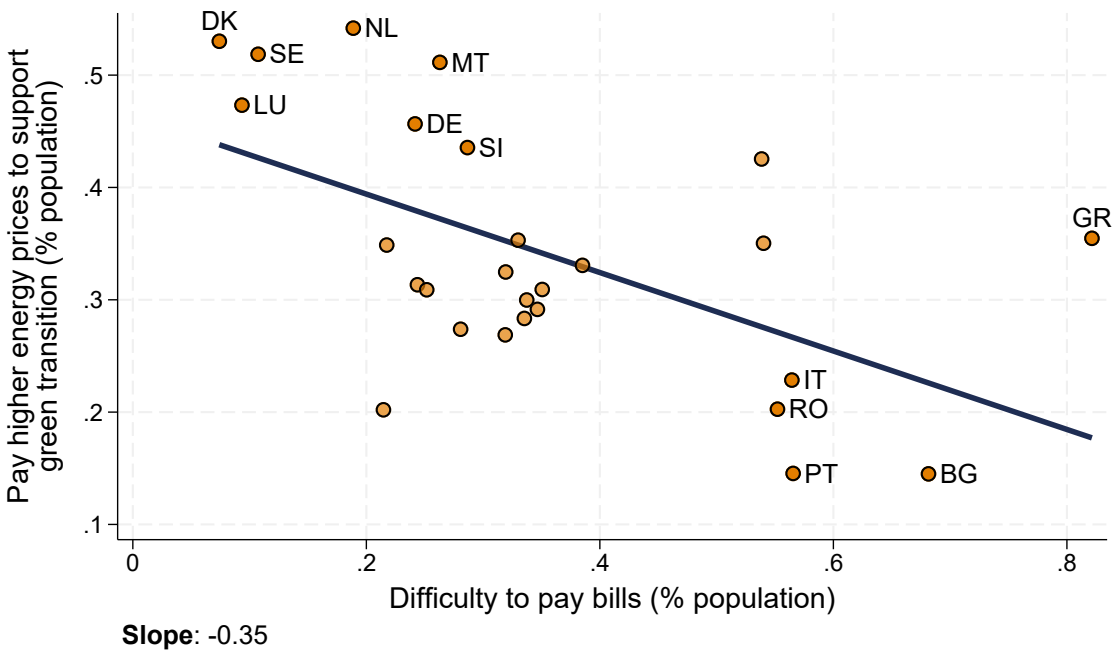

Figure 2: Regression: support for higher energy prices to favour green transition and financial difficulties (% population).

Notes: the figure shows the cross-country correlation between the percentage of people facing difficulties in paying their bills and the percentage of people willing to pay higher energy prices to favour a green transition in 2022. Correlation: -0.349 (p-value<0.01). Countries: Austria, Belgium, Bulgaria, Cyprus, Czech Republic, Germany, Denmark, Estonia, Spain, Finland, France, Greece, Croatia, Hungary, Ireland, Italy, Lithuania, Luxembourg, Latvia, Malta, Netherlands, Poland, Portugal, Romania, Sweden, Slovenia, Slovakia.

regression line), suggesting a distinction between economic and non-economic (or ideological) motive behind the lack of support for climate policies. Starting from the distinction between economic and ideological motives highlighted in Figure 3, in the next section, we present a voting model in which voters with heterogeneous environmental ideologies (ideological motive) and who face uncertainty about their economic conditions (economic motive) vote on a Pigouvian environmental tax to reduce pollution.

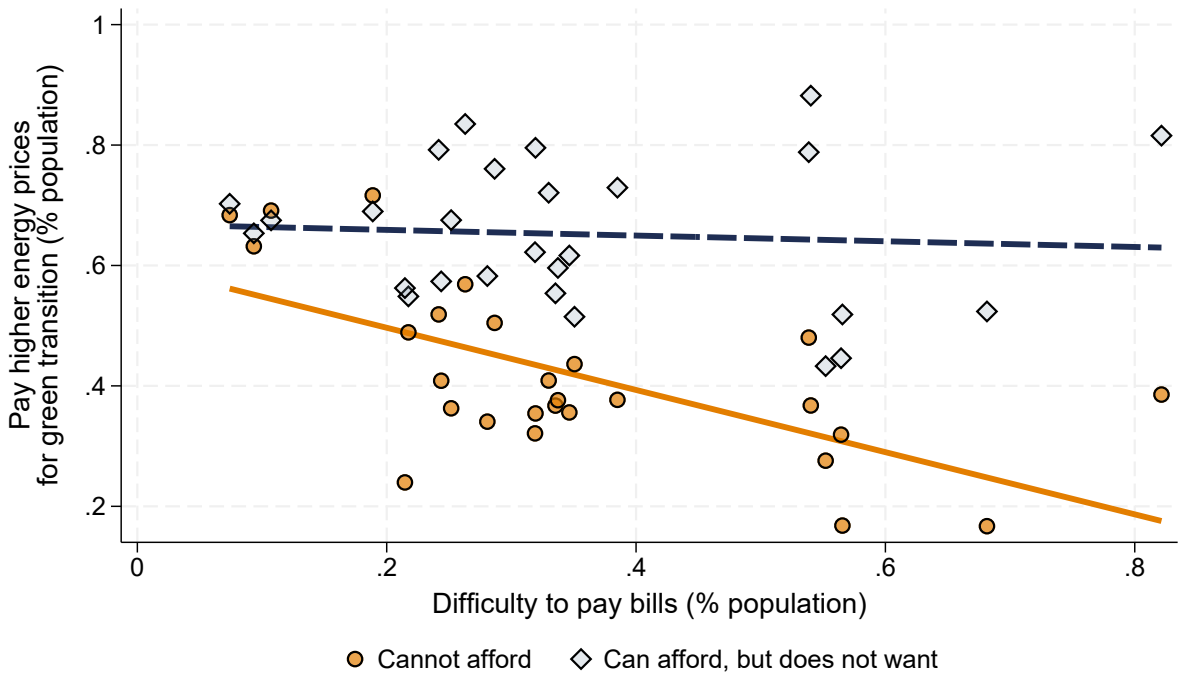

Figure 3: Regression: support for higher energy prices to favour green transition on financial difficulties by type of answer "No" (% population).

Notes: the figure shows the cross-country regression of the average support for higher prices to favour a green transition and the percentage of people not able to pay bills. The blue "diamond" scatterplot (correlation: -0.047; p-value=0.728) represents the regression of the average support for higher prices to favour a green transition when the negative answer is "*No, you are not willing to pay more even though you could afford to do so*". The orange "dot" scatterplot (correlation: -0.517; p-value<0.01) represents the regression of the average support for higher prices to favour a green transition when the negative answer is "*No, you are not willing to pay more as you cannot afford to do so*".

3 The benchmark model

3.1 The economy

We consider an economy producing only one exogenous private good, $q > 0$, whose market price is normalised to 1.[17] Following Fredriksson and Gaston (1999), we assume that the production of one unit of q generates $\alpha \geq 0$ units of pollution, where α captures the damages

[17]In this assumption, we follow Habla and Roeder (2013) and Habla and Roeder (2017) as we build a partial equilibrium model with only one production good which refrains from long-term changes in real wages.

of pollution per unit of q. The higher α is, the more the private good q is polluting. If $\alpha = 0$, the production of q does not generate any damaging pollution and q is referred to as a 'green good'.

The population in the economy consists of a continuum of voters whose size is normalised to 1. At the moment of voting, voters face uncertainty about their future stream of income, so they can be low-income voters with probability $\pi \in (0, 1)$ and high-income voters with probability $1 - \pi$. Low-income voters earn a stream of income $\underline{y} \in [0, \overline{y})$, while high-income voters earn a stream of income $\overline{y} > 0$. As a result, voters face two possible budget constraints: if a voter is low-income, their budget constraint becomes $p(\tau)q = \underline{y}$, where $p(\tau)$ is the post-tax price of the private good, q. On the other hand, if a voter is high-income, their budget constraint becomes $p(\tau)q = \overline{y}$.

Each voter is endowed with a set of ideological values about climate change.[18] Voters' ideology includes multifaceted environmental values and beliefs, such as voters' environmental concerns, pro-climate values or awareness about the causes of climate change (e.g., natural versus anthropogenic). The individual ideology towards climate change, λ_i, is distributed according to a cumulative distribution function, $F(\lambda_i)$, and a probability density function, $f(\lambda_i)$, over support $\left[\underline{\lambda}, \overline{\lambda}\right] \subset \mathbb{R}_{++}$, with a mean $\lambda_a = \int_{\underline{\lambda}}^{\overline{\lambda}} \lambda_i dF(\lambda_i)$ and a median $\lambda_m = F^{-1}(0.5)$. Moreover, individual ideology is assumed to be independent of people's income.[19] Henceforth, for simplicity, we will refer to voters with stronger 'pro-environmental' ideology (or weaker 'anti-environmental' ideology) if λ_i tends to the lower-bound $\underline{\lambda}$ and with stronger 'anti-environmental' ideology (or weaker 'pro-environmental' ideology) if λ_i tends to the

[18] Ideology is a strong predictor of support for climate policies (e.g., Anderson et al. (2023) for evidence from the U.S., Harring and Jagers (2013) for evidence in Sweden and Andre et al. (2024) for worldwide evidence). However, Dechezleprêtre et al. (2022) also show that only some ideological dimensions — such as the anthropogenic view or concerns about the negative consequences of climate change— represent robust predictors.

[19] Although a positive correlation between GDP per capita and awareness about climate change has been found (e.g., see Baiardi and Morana (2021) for Europe), survey data do not show a robust correlation between attitude towards climate change and personal income. For instance, in France, income is unrelated to attitude towards climate change (Douenne and Fabre, 2020, Table 1, p. 8). Moreover, the relationship between personal income and knowledge about climate change is statistically weak or insignificant in most of the 20 countries producing most of the global CO_2 emissions (Dechezleprêtre et al., 2022, Tables A2-A3, pp. 81-82). Also, our stylised facts show a distinction between ideology and economic motive.

upper-bound $\overline{\lambda}$.

3.2 Government and environmental policy

A government decides to introduce an environmental policy to reduce the amount of pollution in the economy if $\alpha > 0$. To this end, the government plans to raise an environmental tax, $\tau \geq 0$, for each unit of emission, such that the post-tax price of one unit of q becomes:[20]

$$p(\tau) = 1 + \alpha\tau \tag{1}$$

The government also reinvests all the revenue from the environmental tax, τ, to subsidise a green good, $g(\tau)$, as a refunding mechanism, such that the government budget constraint becomes:

$$g(\tau) = \tau E(\tau) = \tau\alpha Q(\tau), \tag{2}$$

whereby $E(\tau) = \alpha Q(\tau)$ represents the total emissions in the economy, while $Q(\tau) > 0$ is the aggregate consumption in the economy. Because voters consume all their income, the economy's aggregate consumption and production are the same, such that we can define the aggregate consumption (or production) as a function of the environmental tax, τ:

$$Q(\tau) = \frac{\overline{y}(1-\pi) + \pi\underline{y}}{p(\tau)} \tag{3}$$

Notice that Eq. (3) has some basic economic implications. First, if the environmental tax, τ, increases, the aggregate consumption decreases, thereby showing the distortionary effect of the carbon tax on aggregate consumption. Second, when aggregate consumption decreases because of a higher τ, the total emissions, $E(\tau)$, fall as a result of the environmental policy. Third, as the economy grows (i.e. a higher $\overline{y}$ or $\underline{y}$, or a lower π), the aggregate consumption and the total emissions increase.

[20]For simplicity and without loss of generality (W.l.o.g), we assume there is no other tax on consumption.

3.3 Voters' preferences

Voters' preferences are represented by the expected indirect utility function (4), which depends on the expected utility from private consumption, on the green public good, $g(\tau)$, on the damage from total emissions in the economy, $E(\tau)$, and on the individual level of tax aversion, $\lambda_i \tau$:

$$\mathbb{E}[V_i(\tau)] = \pi \left(\frac{\underline{y}}{p(\tau)} \right) + (1 - \pi) \left(\frac{\overline{y}}{p(\tau)} \right) + g(\tau) - \lambda_i \tau - E(\tau), \tag{4}$$

whereby the expected utility from consumption is the sum, weighted by the probability, π, of the individual private consumption if voters are low-income and the individual private consumption if they are high-income.

The utility component, $\lambda_i \tau$, represents voters' disutility from "tax aversion" and depends on two terms: the individual ideology towards climate change, λ_i, and the environmental tax, $\tau \geq 0$. Our definition of tax aversion relies on the definitions provided by Douenne and Fabre (2022) and Andreassen et al. (2024), namely, a general rejection of higher taxes;[21] thus, tax aversion appears directly in voters' preferences as a disutility function stemming from introducing higher taxes to implement an environmental policy and disappears if the policy is not implemented, that is when $\tau = 0$. For a given environmental tax, τ, a stronger 'anti-environmental' ideology (i.e., a higher λ_i) corresponds to a higher marginal tax aversion; contrarily, a stronger 'pro-environmental' ideology (i.e., a lower λ_i) corresponds to a lower marginal tax aversion.[22]

The timing of the events is the following: first, voters are born and endowed with an individual level of environmental ideology, λ_i. Second, voters vote on the environmental tax,

[21] Precisely, Douenne and Fabre (2022, p. 85) define carbon tax aversion as "[...] a 'gut' rejection of a tax (or taxation in general) that influences beliefs about the tax properties such as its effectiveness, fairness, or sameness with an equivalent measure labelled differently". Andreassen et al. (2024, p. 2) provides a similar definition inspired by Douenne and Fabre (2022)'s.

[22] Alternatively, because if $\tau = 0$, then $g(0) = 0$, one can interpret $g(\tau) - \lambda_i \tau$ as the net value of the green public good perceived by an agent with ideology λ_i. For a given τ, when λ_i is higher, the utility associated with the green public good declines. In other words, people with a stronger 'pro-environmental' ideology receive a higher utility from the implementation of green policies.

τ, and decide the optimal level of green public good and pollution in the economy $(g(\tau), E(\tau))$. Third, a share, π, of voters becomes low-income and produce $\underline{y} \in [0, \underline{y})$. The remaining share, $1-\pi$, becomes high-income and produces $\overline{y} > 0$. Finally, the election-winning environmental tax is applied and affects the price of the good, q.

3.4 Demand for environmental policy

Voters choose their preferred environmental tax by maximising the expected indirect utility function (4) over the tax, τ. As a consequence, voters solve the following maximisation problem to decide their preferred level of environmental tax:

$$\max_{\tau \geq 0} \quad \mathbb{E}[V_i(\tau)] \tag{5}$$

The solution of the maximisation problem (5) leads to the preferred environmental tax function, $\tau^*(\lambda_i)$:

$$\tau^*(\lambda_i) = \max\left\{0, \left[\frac{\overline{y}(1-\pi) + \pi\underline{y}}{\lambda_i}\right]^{1/2} - \frac{1}{\alpha}\right\} \tag{6}$$

The preferred environmental tax (5) is a continuous piecewise function, non-continuously differentiable and decreasing in the 'anti-environmental' ideology, λ_i. Voters with a level of environmental ideology greater than or equal to a rejection threshold, $\widetilde{\lambda}$, where $\widetilde{\lambda} = \alpha^2(\overline{y}(1-\pi) + \pi\underline{y}) > 0$, always demand no environmental tax, such that $\tau^*(\lambda_i) = 0$, for every $\lambda_i \geq \widetilde{\lambda}$. On the other hand, voters with a 'pro-environmental' ideology demand higher environmental tax, such that the maximum demanded environmental tax is $\tau^*(\underline{\lambda}) > 0$.[23]

3.5 Comparative statics

In democratic countries, political decisions must be supported by the majority of voters; this principle is referred to as *majority rule*. By the median voter theorem, as voters are distributed only along the dimension of environmental ideology and their preferences are

[23] See the Online Appendix A for a detailed proof of how to derive the demand function (6).

single-peaked in τ, the election-winning tax coincides with the preferred tax of the median voter, λ_m.[24] As a result, while conducting the comparative statics, we focus on the median voter tax, $\tau^*(\lambda_m)$, assuming that $\lambda_m < \widetilde{\lambda}$ to ensure a positive preferred tax.

3.5.1 Income and exposure to emissions damages

We begin our comparative statics by showing how the preferred environmental tax changes when income and exposure to emissions damages increase. To this end, we differentiate the median voter's preferred environmental tax in Eq. (6) with respect to both $\overline{y}$ and $\underline{y}$:

$$\frac{\partial \tau^*(\lambda_m)}{\partial \overline{y}} = \left[\frac{(1-\pi)^2}{4\lambda_m\left(\overline{y}(1-\pi)+\pi\underline{y}\right)}\right]^{1/2} > 0, \tag{7}$$

$$\frac{\partial \tau^*(\lambda_m)}{\partial \underline{y}} = \left[\frac{\pi^2}{4\lambda_m\left(\overline{y}(1-\pi)+\pi\underline{y}\right)}\right]^{1/2} > 0 \tag{8}$$

Proof. See the Online Appendix B.1 ■

Derivatives (7) and (8) show that the preferred environmental tax is increasing with income. This result holds for both low-income and high-income voters and aligns with the strand of empirical literature showing a positive correlation between income and support for environmental policies (e.g., Levi (2021); Bergquist et al. (2022)).

On the other hand, by differentiating Eq. (6) with respect to α, we also show the effect of an increase in the exposure to damages from emissions:

$$\frac{\partial \tau^*(\lambda_m)}{\partial \alpha} = \frac{1}{\alpha^2} > 0 \tag{9}$$

Derivative (9) is positive, thereby showing that the support for environmental policies increases with the exposure to damages from emissions, as supported by recent empirical evidence (e.g., Hoffmann et al. (2022); Andre et al. (2024)).

[24] We show that preferences are single-peaked in τ in the Online Appendix A.

3.5.2 Economic insecurity

The main purpose of our model is to study the effect of economic insecurity on support for climate policies. We define an increase in economic insecurity as an increase in the probability of becoming a low-income voter. Therefore, to study how preferences for climate policies change when economic insecurity increases, we differentiate the median-voter preferred tax, $\tau^*(\lambda_m)$, with respect to the probability, π, and obtain Proposition 1:

Proposition 1 *An increase in economic security reduces the support for environmental tax, τ, demanded by the median voter, λ_m:*

$$\frac{\partial \tau^*(\lambda_m)}{\partial \pi} = -\left[\frac{\left(\overline{y}-\underline{y}\right)^2}{4\lambda_m\left(\overline{y}(1-\pi)+\pi\underline{y}\right)}\right]^{1/2} < 0 \tag{10}$$

Proof. See the Online Appendix B.2 ■

Proposition 1 shows that the preferred environmental tax decreases with the probability of becoming a low-income voter. This result arises because the environmental tax makes the good, q, more expensive. When economic insecurity increases, the probability of becoming a low-income voter also increases. As a response, voters reduce their demand for the environmental tax to avoid buying more costly goods while being more likely to face a lower income and, in turn, to consume less.

We also calculate the cross-partial derivative of Eq. (10) with respect to λ_m to study how more 'anti-environmental' voters react to an increase in economic insecurity, and we obtain Proposition 2:

Proposition 2 *The more 'anti-environmental' the attitudes of the median voter, λ_m, are, the smaller the effect of higher economic insecurity on the support for environmental tax, τ, is:*

$$\frac{\partial^2 \tau^*(\lambda_m)}{\partial \pi \partial \lambda_m} = \left[\frac{\left(\overline{y}-\underline{y}\right)^2}{16\left(\overline{y}(1-\pi)+\pi\underline{y}\right)}\right]^{1/2}\left(\frac{1}{\lambda_m}\right)^{3/2} > 0 \tag{11}$$

Proof. See the Online Appendix B.3 ■

Proposition 2 shows that the effect of an increase in economic insecurity is smaller for people with stronger 'anti-environmental' ideology. In other words, 'anti-environmental' voters oppose the environmental tax more (see the negative slope of the demand in λ_m), but their opposition is mainly driven by ideological rather than economic reasons. When economic insecurity increases, their incentive to shift preferences toward an even lower tax on pollution and higher emissions diminishes.[25] On the other hand, voters who support higher environmental taxes because of their 'pro-environmental' ideology reduce their political support more if economic insecurity increases. In other words, Proposition 2 states the existence of a stronger "greenlash" from 'pro-environmental' voters. This result occurs because, when 'pro-environmental' voters feel economically insecure, they make an about-face and evaluate their economic motive as marginally more important than their ideological beliefs; thus, they have a higher marginal incentive to backlash against the environmental tax.

4 Welfare analysis

We conclude our benchmark model with a welfare analysis. To this end, we compare the environmental tax voted with the support of a majority to the environmental tax chosen by a social planner aiming to maximise the total welfare in the economy. Thus, we assume that the social welfare planner knows the ideology distribution and chooses the optimal tax by maximising the utilitarian social welfare function, $W(\tau) = \int_{\underline{\lambda}}^{\overline{\lambda}} \mathbb{E}[V_i(\tau;\lambda_i)]\ \mathrm{d}F(\lambda_i)$, such that:[26]

$$\max_{\tau \geq 0} \int_{\underline{\lambda}}^{\overline{\lambda}} \mathbb{E}[V_i(\tau;\lambda_i)]\ \mathrm{d}F(\lambda_i), \tag{12}$$

[25] This result holds because the more a voter supports an 'anti-environmental' ideology, the smaller is their marginal gain from opposing a higher τ in response to a rise in economic insecurity. Moreover, it is important to remember that we assume that $\lambda_m < \widetilde{\lambda}$, so that the median-voter preferred environmental tax is positive. Hence, we do not consider voters who chose the corner solution, $\tau = 0$.

[26] This assumption is not far from politicians adapting to the current public opinion (e.g., Sevenans (2021); Walgrave and Soontjens (2023)).

whereby $\mathbb{E}[V_i(\tau;\lambda_i)]$ is the indirect utility function in Eq. (4), which depends on the individual environmental ideology parameter, λ_i.

Because the population is normalised to 1 and the social planner maximises the total welfare in the economy, the social optimum becomes a function of the average environmental ideology in the economy, λ_a, which we write as $\tau_w^*(\lambda_a)$:[27]

$$\tau_w^*(\lambda_a) = \max\left\{0, \left[\frac{\overline{y}(1-\pi)+\pi\underline{y}}{\lambda_a}\right]^{1/2} - \frac{1}{\alpha}\right\} \tag{13}$$

Comparing the welfare function at the social optimum, $\tau_w^*(\lambda_a)$, and at the optimal environmental tax voted with the support of a majority, $\tau^*(\lambda_m)$, we derive Proposition 3:

Proposition 3 *If a positive environmental tax, $\tau_w^* > 0$, maximises the welfare function, $W(\tau)$, the voting process maximises the total welfare in the economy if and only if $\lambda_m = \lambda_a$. Otherwise, two scenarios can occur:*

- *If $\lambda_m > \lambda_a$, the ideology distribution is left-skewed and the median voter tax is lower than the social optimum, such that $\tau_w^*(\lambda_a) > \tau^*(\lambda_m)$ and $W(\tau^*) < W(\tau_w^*)$.*
- *If $\lambda_m < \lambda_a$, the ideology distribution is right-skewed and the median voter tax is higher than the social optimum, such that $\tau_w^*(\lambda_a) < \tau^*(\lambda_m)$ and $W(\tau^*) < W(\tau_w^*)$.*

Proposition 3 shows that if $\lambda_m \neq \lambda_a$, the median voter preferred environmental tax generates a welfare loss. Results of Proposition 3 are graphically represented in Figure 4. The left-hand side of Figure 4 shows that, when the ideology distribution is left-skewed (i.e. $\lambda_m > \lambda_a$), the voted tax is lower than the social optimum and does not maximise the total welfare in the economy. This result occurs as a left-skewed distribution of environmental ideology implies a higher concentration of voters with an 'anti-environmental' ideology, which will underfinance the environmental tax, thereby reducing the total welfare of the economy. On the other hand, the right-hand side of Figure 4 shows that when the

[27] See the Online Appendix C for a proof.

ideology distribution is right-skewed (i.e. $\lambda_m < \lambda_a$), the voted tax is higher than the social optimum, thereby reducing the total welfare of the economy. A right-skewed environmental ideology distribution entails a higher concentration of voters with a 'pro-environmental' ideology, which will overfinance the environmental tax, thereby reducing the overall welfare of the economy. In both scenarios, in a democratic voting equilibrium, ideology can determine an inefficient allocation of resources to environmental policies, such that $W(\tau^*) < W(\tau_w^*)$, making a welfare-maximiser environmental policy more challenging to implement.

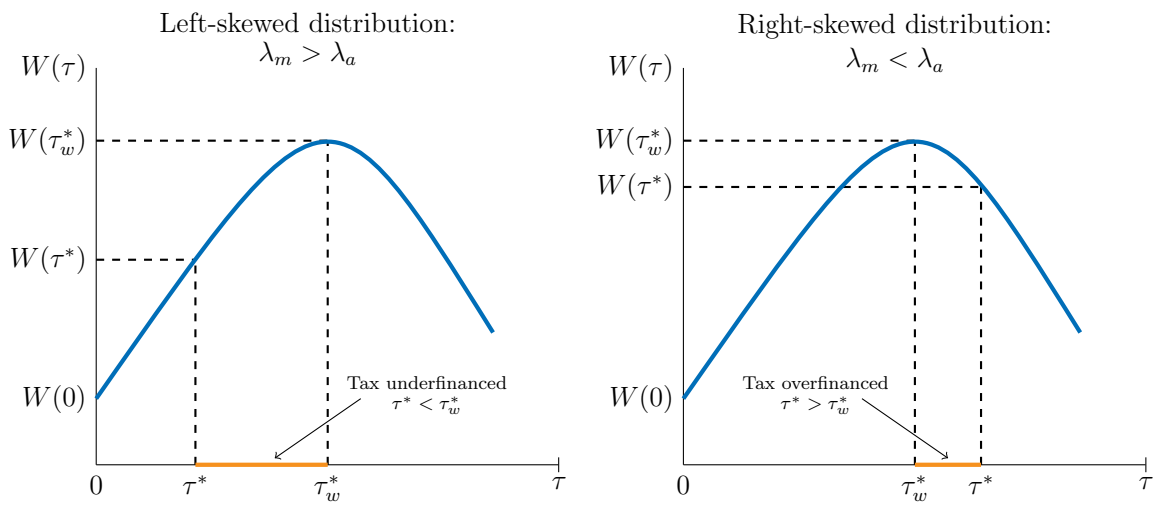

Figure 4: Social optimum, median voter tax and the welfare function

Notes: the graph shows the welfare function computed at the social optimum, τ_w^*, and at the median voter tax, τ^*. On the left-hand graph, the ideology distribution is assumed to be left-skewed, implying an underfunding of the environmental tax. On the right-hand graph, the ideology distribution is assumed to be right-skewed, implying an overfunding of the environmental tax. W.l.o.g, $W(0) > 0$ is assumed.

5 Extensions

In addition to the benchmark model, we also present two extensions explaining some empirical evidence about preferences for climate policies. In our extensions, the role of economic insecurity in shaping preferences for environmental policies remains unchanged; nevertheless, introducing different economic variables provides insights into their potential effects on preferences for environmental policies under income uncertainty.

5.1 Income inequality and social insurance

Empirical evidence shows that income disparities worsen the regressivity of a carbon tax, resulting in a heavier tax burden on low-income individuals compared to high-income individuals (Andersson and Atkinson, 2020). On the other hand, more generous social insurance systems do not affect preferences for environmental policies (Nordbrandt et al., 2024).[28]

In the Online Appendix D, we study how income inequality and the generosity of social insurance systems affect the support for the introduction of an environmental tax. To do so, we adopt a simple definition of pre-tax income inequality, that is the difference between incomes of low-income and high-income voters, denoted as $\Delta = \overline{y} - \underline{y}$. We also assume that the government can reduce income inequality through a "tax-and-transfer" social insurance policy.[29] The social insurance policy entails high-income voters paying a lump sum income tax, t, which is redistributed to all low-income voters in the form of a lump-sum transfer, $b = t(1-\pi)/\pi$, such that $\overline{y} - t > \underline{y} + b$. As a result of the social insurance policy, the post-tax income inequality becomes $\widetilde{\Delta}(t, \pi) = \overline{y} - t - \underline{y} - t(1-\pi)/\pi$, which is strictly decreasing in t. Voters do not vote on the redistributive policy, t, which is exogenously given and exists independently of the implementation of the environmental tax.

The extension shows that if pre-tax income inequality, Δ, increases, the support for climate policies decreases. This result occurs because when income inequality is higher, the environmental tax becomes more regressive, and the tax burden becomes heavier for low-income voters. Indeed, the effect is similar to an income decrease for low-income voters, such that $\partial\tau^*(\lambda_m)/\partial\Delta = -\partial\tau^*(\lambda_m)/\partial\underline{y}$. This result captures recent empirical evidence showing

[28] Precisely, Nordbrandt et al. (2024) show that countries with higher replacement rates and longer duration of social insurance programmes do not affect preferences for climate policies; instead, countries with higher social insurance coverage are more in favour of higher public spending on climate policies. This effect is larger if combined with high replacement rates. In our extension, all the voters are fully covered by the social insurance policy, so the extension isolates only the effect of the generosity of the transfer.

[29] To some extent, in this extension, we also introduce tax progressivity as high-income individuals now have a higher income tax burden than low-income individuals for any given $\tau \geq 0$. Previous research showed that higher tax progressivity affects political and economic decisions as it shifts the tax burden towards high-income individuals (e.g., Doerrenberg and Peichl (2013); Beramendi and Rehm (2016); Gubello and Strecker (2024)). In the context of climate policy, higher tax progressivity (on wealth) can also compensate for the lack of green investments in European countries (Kapeller et al., 2023).

the distributional effect on lower-income households represents one of the main drivers of the lack of support for environmental policies (e.g., Dechezleprêtre et al. (2022)).[30] In addition, we find no effect of more generous social insurance systems — that is, systems with a higher lump-sum transfer, t— on support for environmental policies when income uncertainty is taken into account. This result occurs because the decrease in the tax burden of low-income individuals due to higher monetary transfer, b, is compensated by the increase in the tax burden of high-income individuals due to higher income taxes, t, and voters do not know their future economic status at the moment of the vote.

5.2 Government inefficiency

Political mistrust, perceived government inefficiency and concerns about the misallocation of tax revenue from the carbon tax reduce the support for climate policies (e.g., Hammar and Jagers (2006); Harring and Jagers (2013); Rafaty (2018); Umit and Schaffer (2020); Davidovic et al. (2020)). In the Online Appendix E, we build an extension in which a fraction $\gamma \in [0, 1)$ of the tax revenue from the environmental tax is wasted and not turned into a valuable green public good, $g(\tau)$, due to government's inefficiency. The extension shows two main results: first, a positive demand for the environmental tax, $\tau^*(\lambda_m) > 0$, can exist only if the marginal damage of an extra unit of pollution, α, is greater than the marginal waste of resources, γ; otherwise voters will always choose $\tau^*(\lambda_i) = 0$. Second, more inefficient public institutions reduce the support of the environmental tax and intensify the negative effect of higher economic insecurity on support for climate policies.

6 Data

In the next sections, we bring our benchmark model to the data. Although causality cannot be claimed, our empirical analysis provides robust descriptive evidence among European

[30] The response to inequality is also termed as "inequality concerns" (Dechezleprêtre et al., 2022).

countries supporting the predictions of the theoretical model. Hence, we create a new dataset by merging wave 8 of the European Social Survey (henceforth, ESS) (European Social Survey (ESS 8), 2023), which provides individual-level data, with the Quality of Government (henceforth, QoG) EU regional dataset (Charron et al., 2020), which provide region-level data. The final dataset consists 18,604 respondents from 17 European countries and 160 regions. The countries are Belgium, Czech Republic, Germany, Spain, Finland, France, United Kingdom, Hungary, Ireland, Italy, Lithuania, Netherlands, Norway, Poland, Portugal, Sweden, and Slovenia. Detailed descriptive statistics can be found in the Online Appendix H.

6.1 Individual data

The individual-level data comes from wave 8 of the ESS (European Social Survey (ESS 8), 2023).[31] Wave 8 of the ESS covers 23 countries over the years 2016-2017 and presents questions about welfare attitudes and attitudes towards climate change. The ESS provides information about the respondents' regions of residence. Regional information covers different NUTS (*Nomenclature of Territorial Units for Statistics*) levels, depending on the country. We re-aggregated some regions to merge the ESS with the QoG EU regional dataset. Full information about the aggregation and a final sample of regions used in this analysis can be found in the Online Appendix G.

6.1.1 Environmental policy support

The main dependent variable in our analysis is the level of support for environmental policies. In our theoretical model, a government implements a carbon tax to subsidise a green public good. To adhere closely to the assumptions of the model, we construct a principal component analysis (PCA) index using two questions following the same structure: "*To what extent are you in favour or against the following policies in [country] to reduce climate change?*": (i) "*Increasing taxes on fossil fuels, such as oil, gas and coal*", taking values going from

[31] See the ESS webpage: `https://www.europeansocialsurvey.org/about-ess`.

1 ("*Strongly in favour*") to 5 ("*Strongly against*"); (ii) "*Using public money to subsidise renewable energy such as wind and solar power*", taking values going from 1 ("*Strongly in favour*") to 5 ("*Strongly against*"). These two questions imply introducing a carbon tax and green energy subsidies to reduce climate change in line with the theoretical model assumptions. We reverse the values of the answers such that higher values are associated with higher support for green policies. The first principal component explains 61.66% of the total variance, and the eigenvalue of the principal component equals 1.233.[32] The higher the PCA index is, the higher is the individual support for environmental policies. We normalised the index to take values in the interval $[0, 100]$ to simplify the empirical interpretation.

6.1.2 Individual economic insecurity

Our measure of perceived individual economic insecurity relies on Bossert et al. (2023)'s measure. The authors exploit longitudinal data to create a proxy for individual economic insecurity, which reflects an individual's confidence in facing future economic challenges and is affected by the income variations an individual has experienced throughout their life. Because our data are not longitudinal, we use the first principal component of a PCA capturing perceived economic insecurity. Using a PCA allows us to align with the economic insecurity definition by Bossert et al. (2023) by combining variables about past experiences of unemployment and concerns about the current and future financial situation. Hence, we use three questions from wave 8 of the ESS: (i) "*During the next 12 months how likely is it that there will be some periods when you don't have enough money to cover your household necessities?*", taking values going from 1 ("*Not at all likely*") to 4 ("*Very likely*"); (ii) "*Have you ever been unemployed and seeking work for a period of more than three months?*", taking values 1 ("*Yes*") or 0 ("*No*"); (iii) "*Which of the descriptions on this card comes closest to how you feel about your household's income nowadays?*", taking values going from 1 ("*Living comfortably on present income*") to 4 ("*Very difficult on present income*"). The first prin-

[32] See Online Appendix F.1 for detailed results of the PCA analysis.

cipal component explains 55.92% of the total variance and the eigenvalue of the principal component equals 1.678.[33] We normalised the index such that it takes values in the interval $[0, 100]$. The higher the PCA index is, the higher is the individual level of economic insecurity.

6.2 Regional data

The aggregate regional data comes from the QoG EU regional dataset (Charron et al., 2020). The QoG EU regional dataset includes several European regional aggregate variables at different NUTS levels. From this dataset, we collect regional information about population density per km^2, total fertility rate, log of GDP per capita, percentage of population with tertiary education and long-term unemployment as a percentage of total unemployment. We rely on long-term unemployment to account for persistent long-term structural issues in the regional labour market. Moreover, we build a new variable consisting of the ratio between the percentage of employment in agriculture and manufacturing and the percentage of employment in services to measure the composition of the regional labour force. Henceforth, we referred to this ratio as the "blue-white collar employment ratio".

7 Empirical analysis

In this section, we discuss empirical evidence supporting the model's predictions on the relationship between perceived economic insecurity and support for welfare policies. We test the relationship between support for environmental policies and economic insecurity by estimating the following fixed-effect OLS regression model:[34]

$$EnvPolSup_{irt} = \beta_1 Insecurity_{irt} + \boldsymbol{X'}_{irt}\boldsymbol{\gamma} + \boldsymbol{Z'}_{rt}\boldsymbol{\vartheta} + \mu_c + \theta_t + \varepsilon_{irt}, \tag{14}$$

where $EnvPolSup_{irt}$ is the continuous measure of support for environmental policies stan-

[33] See Online Appendix F.2 for detailed results of the PCA analysis.

[34] In the Online Appendix I, we also conduct a bivariate analysis providing preliminary evidence of the negative relationship between support for climate policies and economic insecurity by country.

dardised in the interval $[0, 100]$ of an individual i in the region r at time t, $Insecurity_{irt}$ is the measure of perceived economic insecurity for an individual i in the region r at time t. $\mathbf{X}_{irt}$ is a set of individual controls (income, occupation, employment status), socioeconomic controls (gender, education, age, household size, residence, marital status and citizenship) and ideology controls (religiosity, left-right scale political orientation, trust in politicians, climate change attitudes). The variable capturing climate change attitudes is the first principal component of a PCA and takes values in $[0, 100]$. The higher the PCA index is, the higher is the individual level of environmental concern and climate change awareness.[35] $\mathbf{Z}_{rt}$ is the set of regional controls listed in Section 6.2. Finally, we include country fixed-effect, μ_c, and year fixed-effect, θ_t, to control for unobserved heterogeneity across countries and over time. Standard errors are clustered at the region level.

7.1 Main results

Table 1 shows the correlation between individual economic insecurity and support for climate policies. In Column (1), we control only for economic insecurity. Column (1) shows that a 1 percentage point (pp) increase in the economic insecurity scale reduces the support for climate policies by 0.088 pp. The result is statistically significant at the 1% level, thereby confirming Proposition 1. In Column (2), we introduce the regional aggregate variables, and a 1 pp increase in the economic insecurity scale reduces the support for climate policies by 0.081 pp. In Column (2), the results remain statistically significant at the 1% level.

In Columns (3) and (4), we introduce income and other individual controls. The magnitude of economic insecurity halves such that a 1 pp increase in the economic insecurity scale reduces the support for climate policies by 0.041 pp. Yet, the correlation remains statistically significant at the 1% level. On the other hand, the income variable is positively correlated with support for climate policies and statistically significant at the 1% level in Column (3); however, in Column (4) the statistical significance of income decreases at the

[35] A full description of all the control variables can be found in the Online Appendix H.

5% level. In Column (5), when we add the ideology controls, we can observe that economic insecurity remains stable and negatively correlated with the support for climate policies at the 1% level. Indeed, a 1 pp increase in the economic insecurity scale reduces the support for climate policies by 0.046 pp. Instead, climate change attitudes are positively correlated with the support for climate change policies at the 1% level, while income becomes statistically insignificant. This outcome aligns with the empirical evidence which finds a positive but statistically weak correlation between income and willingness to support climate policies.

Table 1: Regression: Support for environmental policies and perceived economic insecurity

	Environmental policies support				
Variables	(1)	(2)	(3)	(4)	(5)
Economic insecurity	-0.088***	-0.081***	-0.041***	-0.041***	-0.046***
	(0.012)	(0.011)	(0.012)	(0.012)	(0.011)
Income			0.313***	0.256**	0.141
			(0.100)	(0.099)	(0.090)
Climate change attitudes					0.391***
					(0.015)
Country-Year FE	✓	✓	✓	✓	✓
Regional Controls		✓	✓	✓	✓
Income Controls			✓	✓	✓
Individual Controls				✓	✓
Ideology Controls					✓
Regions	160	160	160	160	160
Observations	18,604	18,604	18,604	18,604	18,604
Adjusted R-squared	0.061	0.068	0.091	0.100	0.192

Notes: Columns (1)-(5) report the support for environmental policies regressed on perceived economic insecurity (1), aggregate regional controls (2), income controls (3), individual controls (4) and ideology controls (5). Country- and year-fixed effects are always included. Standard errors clustered at the regional level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

7.2 Cost channels

The model shows that voters react to an increase in economic insecurity by reducing their support for climate policies. If they become low-income, they are more likely to face a heavier tax burden as their income decreases; thus, the main reason why support for the environmental tax decreases is the concern for higher costs and lower purchasing power.

To test whether the concern about higher costs is the main channel of the negative effect of economic insecurity on the support for climate policies, we present the same specification as in model (14) using a different set of dependent variables. The new dependent variables are based on different ESS questions. To capture the concern about energy costs, we follow Arndt et al. (2023) and use the question (i) *"How worried are you that energy may be too expensive for many people in [country]?"* taking values from 1 ("*Not at all worried*") to 5 ("*Extremely worried*"). In addition, as placebo tests, we use a set of questions that do not entail higher costs. First, we use a question about the implementation of an alternative climate policy which does imply higher consumption taxes: (ii) *"To what extent are you in favour or against the following policies in [country] to reduce climate change? A law banning the sale of the least energy efficient household appliances"*, taking values from 1 ("*Strongly in favour*") to 5 ("*Strongly against*"). Second, we use four questions on the support for four renewable energies (i.e., biomass, wind, solar and hydroelectric power), which do not involve a cost increase. The answers with higher values are associated with higher support for renewable energies.[36]

Table 2 shows that the concern for higher costs is a plausible channel through which people oppose a climate policy involving higher taxes. Indeed, Column (1) shows that higher

[36]The questions about renewable energy all follow the same structure: *"How much of the electricity used in [country] should be generated from each energy source?"* (iii) *"[...] biomass energy?"*; (iv) *"[...] wind power?"*; (v) *"[...] solar power?"*; (vi) *"[...] hydroelectric power?"*, taking values from 1 ("*A very large amount*") to 5 ("*Not at all*"). We reverse the values of questions (ii)–(vi) so that higher values correspond to higher support for the alternative climate policy and stronger preferences for renewable energy. We standardise variables (i)–(vi) with a mean equal to 0 and a standard deviation of 1. There is also an answer *"I have not heard of this energy source before"* taking value 55. We exclude respondents providing this answer. However, they constitute a small minority, ranging from 1.67% (307 observations) for biomass energy to 0.15% (28 observations) for wind power.

economic insecurity and concern for energy costs are positively correlated and statistically significant at the 1% level. On the other hand, economic insecurity is statistically insignificant with all the placebo questions in Columns (2)–(6). Moreover, because climate change ideology is independent of the implementation cost, the variable capturing individual attitude towards climate change is always positively correlated with all the dependent variables and statistically significant at the 1% level.[37]

Table 2: Regression: cost channel and placebo tests

	Cost channel	Alternative policy	Preferences for renewable energy			
	Energy expensive	Ban inefficient energy	Biomass energy	Wind power	Solar power	Hydroelectric power
Variables	(1)	(2)	(3)	(4)	(5)	(6)
Economic insecurity	0.009***	-0.001	0.001	-0.001	0.000	0.000
	(0.001)	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)
Income	0.004	0.003	0.009*	-0.002	-0.000	0.006
	(0.008)	(0.004)	(0.005)	(0.004)	(0.006)	(0.004)
Climate change attitudes	0.006***	0.015***	0.005***	0.011***	0.011***	0.005***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Regions	160	160	160	160	160	160
Observations	18,568	18,604	17,871	18,444	18,468	18,363
Adjusted R-squared	0.238	0.094	0.075	0.158	0.146	0.098

Notes: Columns (1)-(6) show the effect of perceived economic insecurity on (1) concern for energy expenses, (2) preferences for banning the sale of inefficient household appliances, and preferences for alternative energy sources such that (3) biomass energy, (4) wind power, (5) solar power and (6) hydroelectric power. Aggregate regional controls, income controls, individual controls, ideology controls and country- and year-fixed effects are always included. Standard errors clustered at the regional level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

7.3 Heterogeneity

In this section, we study if the effect of economic insecurity is heterogeneous across groups of individuals. We provide three heterogeneous analyses: (i) heterogeneity across individual

[37]The correlation is positive also for the "energy cost" dependent variable in Column (1). Although this finding is not new (e.g., see Arndt et al. (2023)), a plausible explanation is that people with a higher awareness of climate change also have a greater awareness of what climate change can imply for energy costs.

socioeconomic characteristics, (ii) heterogeneity across ideological characteristics, and (iii) heterogeneity across regional characteristics.

7.3.1 Socioeconomic heterogeneity

Table 3 shows the heterogeneous effect by different socioeconomic categories, that is income levels (i.e., bottom quintile, top quintile and people in between as reference category), higher education, people living in a big city or suburb of a big city, people older than 65 and male. We also add in the model a variable capturing if people have children at home, as the "legacy hypothesis" (e.g., see Thomas et al. (2018)) suggests that people with children could care more about the future of the environment. Most of the categories are not statistically significant;[38] the only two categories that are statistically significant — and only at the 10% level— are: male, with a negative effect of economic insecurity 60% smaller than people younger than female, and people over 65, with a negative effect of economic insecurity 12% larger than people younger than 65. Interestingly, the interaction between income and economic insecurity is statistically insignificant, further supporting the argument by Bossert et al. (2023) for which economic insecurity is not related to the income level of people.

7.3.2 Political and risk attitudes heterogeneity

Table 4 shows the heterogeneous effect across different political attitudes, namely respondents with high institutional trust, left-centrist-right scale, and respondents with 'pro-environmental' attitudes. Moreover, we add two dummy variables capturing individual risk aversion and prudence.[39] Results are presented in Table 4. The main finding here is that

[38]The lack of effect for those with children is also found in Thomas et al. (2018).

[39]The dummy variables are constructed as follows: we define the reference category "Centrists" as those respondents who rated themselves as 5 on the 0–10 left-right scale, where 0 corresponds to "*Left*" and 10 to "*Right*". We define "high institutional trust" as those respondents who rated themselves as having trust in the parliament greater or equal to 5 (the median), where 0 corresponds to "*No trust at all*" and 10 to "*Complete trust*". We define 'pro-environmental' attitudes as those respondents reporting the environmental attitudes index above the median. The ESS questions for the variables "Risk aversion" and "Prudence" are described in the Online Appendix J. We create two dummy variables, assigning value 1 if respondents select answers between 1 ("*Very much like me*") and 3 ("*Somewhat like me*") and 0 otherwise. The variable for "Risk aversion" has already been used in Guiso et al. (2024).

Table 3: Regression: heterogeneity with individual socio-demographic controls

	Environmental policies support					
Variables	(1)	(2)	(3)	(4)	(5)	(6)
Economic insecurity	-0.046***	-0.045***	-0.052***	-0.039***	-0.056***	-0.042***
	(0.013)	(0.012)	(0.013)	(0.012)	(0.012)	(0.012)
Econ. insecurity × Bottom quintile	-0.000					
	(0.023)					
Econ. insecurity × Top quintile	0.017					
	(0.020)					
Econ. insecurity × Higher education		-0.007				
		(0.018)				
Econ. insecurity × City			0.024			
			(0.016)			
Econ. insecurity × Over 65				-0.044*		
				(0.024)		
Econ. insecurity × Male					0.022*	
					(0.013)	
Econ. insecurity × Children at home						-0.010
						(0.014)
Regions	160	160	160	160	160	160
Observations	18,604	18,604	18,604	18,604	18,604	18,604
Adjusted R-squared	0.192	0.192	0.191	0.192	0.192	0.192

Notes: Columns (1)-(6) show the interaction effect between economic insecurity and income (1), higher education (2), living in a city (3), being over 65 (4), being a male (5) and having children at home (6). Aggregate regional controls, income controls, individual controls, ideology controls and country- and year-fixed effects are always included. Each variable interacting with perceived economic insecurity is included as a control in the regression but not shown in the table. Standard errors clustered at the regional level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

the negative impact of economic insecurity on support for environmental policies is driven by those with 'pro-environmental attitudes', which is in line with Proposition 2. No heterogeneity is found based on being left-wing, risk-averse or prudent. Those with higher trust in institutions are more likely to support environmental taxes, which is not surprising given the literature discussed earlier regarding political mistrust and perceived government inefficiency, but this is only significant at the 10% level.

Table 4: Regression: heterogeneity with political and risk attitude controls

	Environmental policies support				
	Political attitudes			Risk attitudes	
Variables	(1)	(2)	(3)	(4)	(5)
Economic insecurity	-0.057***	-0.064***	-0.022	-0.048***	-0.042***
	(0.017)	(0.014)	(0.015)	(0.013)	(0.012)
Economic insecurity × Right-wing	0.044**				
	(0.018)				
Economic insecurity × Left-wing	-0.008				
	(0.019)				
Economic insecurity × High institutional trust		0.027*			
		(0.014)			
Economic insecurity × Pro-environmental attitudes			-0.036**		
			(0.016)		
Economic insecurity × Risk averse				0.002	
				(0.015)	
Economic insecurity × Prudent					-0.022
					(0.018)
Regions	160	160	160	160	160
Observations	18,604	18,604	18,604	18,489	18,502
Adjusted R-squared	0.192	0.188	0.163	0.193	0.193

Notes: Columns (1)-(5) show the interaction effect between economic insecurity and (i) political attitudes (right-left ideology (1), institutional trust (2) and 'pro-environmental' attitudes (3)); (ii) risk attitudes (risk aversion (4) and prudence (5)). Aggregate regional controls, income controls, individual controls, ideology controls and country- and year-fixed effects are always included. Each variable interacting with perceived economic insecurity is included as a control in the regression but not shown in the table. Standard errors clustered at the regional level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

7.3.3 Regional heterogeneity

Table 5 shows the heterogeneous effect of different regional economic characteristics of the regions of residence of the respondents in the ESS. We conduct a heterogeneous analysis distinguishing between regions with low and high long-term unemployment, blue-white employment ratio, share of tertiary educated people, and GDP per capita. We define a region with high long-term unemployment, high blue-white employment ratio, high share of tertiary educated people, or high GDP per capita and assign a value of 1 if the value for the specific

variable in that region is above the median sample. Otherwise, we assign a value of 0. No heterogeneous effects are found apart from long-term unemployment — here we see that the effect of economic insecurity on support for environmental taxes is entirely driven by those in regions with high levels of long-term unemployment. This may reflect the fact that people in such regions are more acutely aware of economic conditions and feel they have higher exposure to economic insecurity.[40]

7.4 Robustness checks

We conduct several robustness checks:

Leave-one-country-out. We estimate each coefficient by estimating the OLS regression (14), each time leaving one different country out of the sample. Figure K.1 in the Online Appendix shows that economic insecurity is always statistically significant, with the coefficients having similar sizes, thereby excluding the presence of country-specific trends.

Income as predictor and non-linear income effect. In Table K.3 of the Online Appendix, we run the model (14) using income deciles as a categorical variable to test income non-linearity. In Table K.4, we also replicate Table 1 without using the income control (Panel (a)) and without using the economic insecurity control (Panel (b)). Both control variables are statistically significant at the 1% level when the other control is excluded, suggesting that perceived economic insecurity is a stronger predictor of preferences for environmental policies.

Cost channel: carbon tax vs green subsidy. In Table K.5 of the Online Appendix,

[40] In Table K.1 of the Online Appendix, we also interact the regional variables with the attitudes toward climate change. The interaction is positive and statistically significant at the 1% and 5% level for regions with high share of tertiary educated people and high GDP per capita, respectively. This result aligns with the findings of a correlation between GDP per capita and 'pro-environmental' attitudes (e.g., Baiardi and Morana (2021)). On the other hand, the interaction is negative for regions with high long-term unemployment and high blue-white employment ratio (5% and 1% levels, respectively). The climate change attitude control remains statistically significant at the 1% level, meaning that, while these regional heterogeneous effects exist, they do not drive the results of individual attitudes. Moreover, in Table K.2, we interact the regional long-term unemployment variable with the dummy for 'pro-environmental' individuals, like in Table 4. We find that the negative effect of regional unemployment is driven by 'pro-environmental' respondents, showing their stronger "greenlash" also when using aggregate measures of economic insecurity.

Table 5: Regression: heterogeneity with regional controls

Variables	Environmental policies support (1)	(2)	(3)	(4)
Economic insecurity	-0.021	-0.054***	-0.045***	-0.035**
	(0.013)	(0.014)	(0.017)	(0.018)
Econ. insecurity × High long-term unemployment	-0.045**			
	(0.018)			
Econ. insecurity × High blue-white collar employment ratio		0.015		
		(0.020)		
Econ. insecurity × High share of tertiary educated			-0.001	
			(0.020)	
Econ. insecurity × High GDP per capita				-0.020
				(0.020)
Regions	160	160	160	160
Observations	18,604	18,604	18,604	18,604
Adjusted R-squared	0.192	0.192	0.192	0.192

Notes: Columns (1)-(4) show the interaction effect between economic insecurity and regions with high long-term unemployment (1), a high blue-white collar employment ratio (2), a high share of tertiary educated people (3) and a high GDP per capita (4). Aggregate regional controls, income controls, individual controls, ideology controls and country- and year-fixed effects are always included. Each variable interacting with perceived economic insecurity is included as a control in the regression but not shown in the table. Standard errors clustered at the regional level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

we also run the model (14) using the two questions about the carbon tax and the green subsidy separately as dependent variables. We use an OLS specification and an ordered probit specification. In both econometric specifications, perceived economic insecurity is negatively correlated and statistically significant at the 1% level when the question about the carbon tax is used as the dependent variable. Instead, the statistical significance shrinks at the 5% level for the OLS specification and the 10% level for the ordered probit specification with the green subsidy question. Instead, the statistical significance of climate change attitudes does not change. This result confirms that the higher expected cost represents the main channel through which economic insecurity affects preferences for climate policies.

Cost channel: redistributive policies. In Table K.6 of the Online Appendix, we replicate regression (14) using as dependent variables two questions about preferences for

two specific redistributive policies (i.e., basic income and EU social benefit scheme) and two questions about preferences for redistribution and inequality reduction. If higher costs and deteriorated purchasing power represent the main channel for opposing climate policies, we should expect people with higher economic insecurity to show stronger support for redistributive policies as these policies aim to cushion the negative effects of income losses. Table K.6 shows that respondents with higher economic insecurity show greater support for the two redistributive policies and stronger preferences for redistribution, further supporting the cost channel hypothesis.

8 Discussion and conclusion

While targets exist for the reduction of carbon emissions, countries continue to face challenges in implementing related policies due to the lack of voters' support (e.g., Drews and Van den Bergh (2016)). This paper has examined the link between perceived economic insecurity and support for climate policies. The main finding is that perceived economic insecurity is negatively correlated with support for policies to tackle climate change. Once climate change attitudes are taken into account, we find no link between income and support for environmental taxes, while the negative correlation with perceived economic insecurity remains. No strong heterogeneous effects are found across different socioeconomic characteristics. In terms of regional heterogeneity, the effect is driven by those in regions with high long-term unemployment. In line with the theoretical model, the negative correlation between perceived economic insecurity and support for environmental taxes is driven by respondents with 'pro-environmental' attitudes. In other words, those with 'pro-environmental' attitudes reduce their support for environmental taxes more than those with 'anti-environmental' attitudes as perceived economic insecurity rises.

These findings have important implications for policymakers. While the policy focus is often on the stronger negative impact of environmental taxes on those with lower incomes

and how policy can help target support to this group (e.g., Eurofound (2021); ILO-OECD (2021)) — often with the intention of making environmental taxes more palatable to the electorate — our findings suggest that income has no effect on support for environmental taxes once ideology has been controlled for. This does not mean that the stronger effect of policies like carbon taxes on lower-income groups should be ignored, but rather that targeted support may do little to increase voter support for environmental taxation.

Our findings instead suggest that perceived economic insecurity is a key driver of the support for environmental taxes, independent of income. The effect is particularly driven by those who live in a region with high long-term unemployment. Economic insecurity has been on the rise over the last few decades, exacerbated recently by the COVID pandemic and cost-of-living crises, which have eroded purchasing power and led to decreased savings and increased debt (OECD, 2023a). While economic fragility is more prevalent at the bottom end of the income distribution, it is not restricted to it, with the OECD finding that while two in three lower-income households are financially fragile, so too are nearly half of middle-income households (OECD, 2023a).[41] Economic insecurity is likely to rise in the coming years due to AI — especially if the "productivity gains" from these new technologies are not equally distributed across the society (e.g., see Johnson and Acemoglu (2023)) —, particularly for roles needing higher-level cognitive skills that have a high risk of automation (Lassébie and Quintini, 2022) and medium-paying, "routine-clerical jobs" (Simons et al., 2024).

Therefore, policies that help reduce economic insecurity and long-term unemployment would seem to be key in also tackling the environmental crisis and driving support for environmental taxes. A suite of policies would be needed, and a full discussion is beyond the scope of this paper; however, OECD (2023a) discusses a range of policies to reduce economic insecurity, such as macroeconomic policies that help maintain strong economic conditions and invest in the skills and programmes needed to support the digital and green transition along with micro-level policies such as improving savings schemes, financial literacy and re-

[41] A person is deemed to be 'financial fragile' if "they do not have sufficient liquid assets to stay above the poverty line for three months" (OECD, 2023a, p. 3).

ducing individual level over-indebtedness. Our findings suggest that such policies would also help increase voter support for measures needed to tackle the climate crisis.

We provided descriptive evidence — using different data sources— of the robust negative correlation between perceived economic insecurity and support for Pigouvian climate policies. Future research should focus on estimating a causal effect of the perception of economic insecurity and support for climate policies. Furthermore, it would be relevant for academic and policy research to test if the larger "greenlash" effect on people with stronger 'pro-environmental' attitudes is also observed outside of European countries. A more detailed study of the psychological and sociological mechanisms behind this stronger backlash from 'pro-environmental' citizens would also benefit policymakers, as it could provide a better understanding of designing climate policies and help achieve a cross-national green transition.

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Economic insecurity, greenlash and the demand for climate policies in Europe

(ONLINE APPENDIX — Supplementary material)

A Agents' preferences

A.1 Proof: Single-peaked preferences

To prove that preferences are single-peaked in τ, we need to show that $\frac{\partial^2 V(\tau)}{\partial^2 \tau} < 0$. Consider the indirect utility function:

$$\mathbb{E}[V_i(\tau)] = \frac{\overline{y}(1-\pi)+\pi\underline{y}}{1+\alpha\tau} + \tau\alpha\frac{\overline{y}(1-\pi)+\pi\underline{y}}{1+\alpha\tau} - \alpha\frac{\overline{y}(1-\pi)+\pi\underline{y}}{1+\alpha\tau} - \tau\lambda_i \tag{A.1.1}$$

The FOC are:

$$\frac{\partial\mathbb{E}[V_i(\tau)]}{\partial\tau} : \ \frac{\alpha^2(\overline{y}(1-\pi)+\pi\underline{y})}{(1+\alpha\tau)^2} - \lambda_i \leq 0 \tag{A.1.2}$$

The SOC are:

$$\frac{\partial^2\mathbb{E}[V_i(\tau)]}{\partial^2\tau} : \ -\frac{2\alpha^3(\overline{y}(1-\pi)+\pi\underline{y})}{(1+\alpha\tau)^3} < 0 \tag{A.1.3}$$

Because the function $V(\tau)$ is strictly concave in τ, there must exist a unique τ^* that maximise the function.

A.2 Maximisation problem

Agents solve the following maximisation problem:

$$\max_{\tau\geq 0} \quad \frac{\overline{y}(1-\pi)+\pi\underline{y}}{1+\alpha\tau} + \tau\alpha\frac{\overline{y}(1-\pi)+\pi\underline{y}}{1+\alpha\tau} - \alpha\frac{(\overline{y}(1-\pi)+\pi\underline{y})}{1+\alpha\tau} - \tau\lambda_i \tag{A.2.1}$$

To solve the maximisation problem, we apply the Lagrangian method. The Lagrangian function is:

$$\mathcal{L} = \frac{\overline{y}(1-\pi)+\pi\underline{y}}{1+\alpha\tau} + \tau\alpha\frac{\overline{y}(1-\pi)+\pi\underline{y}}{1+\alpha\tau} - \alpha\frac{(\overline{y}(1-\pi)+\pi\underline{y})}{1+\alpha\tau} - \tau\lambda_i + \nu_1\tau \tag{A.2.2}$$

where $\nu_1 \geq 0$ is the lagrangian multiplier.

The necessary conditions for a local maximum are given by the FOCs. The FOCs must solve the Kuhn-Tucker conditions: (A.2.3) -(A.2.5):

$$\frac{\partial\mathcal{L}}{\partial\tau} \leq 0: \ \frac{\alpha^2(\overline{y}(1-\pi)+\pi\underline{y})}{(1+\alpha\tau)^2} - \lambda_i + \nu_1 \leq 0, \tag{A.2.3}$$

$$\nu_1\frac{\partial\mathcal{L}}{\partial\nu_1} = 0: \ \nu_1\tau = 0, \tag{A.2.4}$$

$$\nu_1 \geq 0 \tag{A.2.5}$$

Henceforth, we verify the existence of both an interior and a corner solution:

- **Case 1 (corner solution):** $\nu_1 > 0$ and $\tau = 0$.

The Kuhn-Tucker condition (A.2.3), then becomes:

$$\alpha^2(\overline{y}(1-\pi)+\pi\underline{y}) - \lambda_i + \nu_1 \leq 0 \tag{A.2.6}$$

Because $\nu_1 > 0$ (A.2.6) is satisfied only if:

$$\lambda_i \leq \alpha^2(\overline{y}(1-\pi)+\pi\underline{y}) \tag{A.2.7}$$

- **Case 2 (interior solution):** $\nu_1 = 0$ and $\tau > 0$.

An interior solution is found by setting (A.2.3) equal to 0 and solving it for τ:

$$\frac{\alpha^2(\overline{y}(1-\pi)+\pi\underline{y})}{(1+\alpha\tau)^2} - \lambda_i = 0 \tag{A.2.8}$$

which gives the following optimal tax rate $\tau^* > 0$:

$$\tau^* = \left(\frac{\overline{y}(1-\pi)+\pi\underline{y}}{\lambda_i}\right)^{1/2} - \frac{1}{\alpha} \tag{A.2.9}$$

Figure A.1 shows a plot of the demand function:

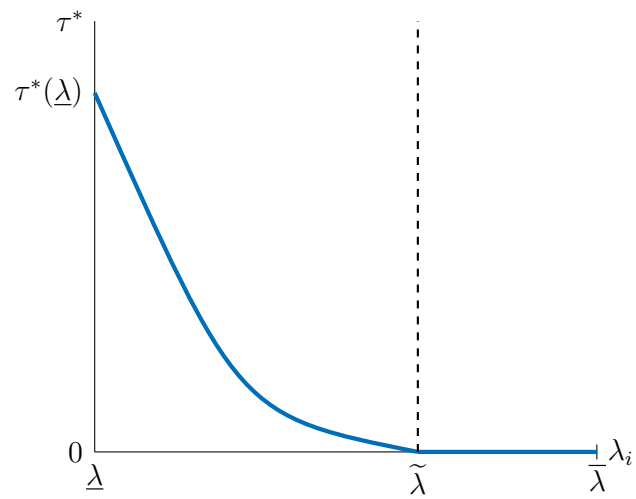

Figure A.1: Demand for environmental tax τ

Notes: the graph shows the demand for environmental tax, τ, as a function of environmental ideology, λ_i. The demand is positive if the environmental ideology is smaller than the no-policy threshold, $\widetilde{\lambda}$. Otherwise, voters demand an environmental tax, $\tau = 0$.

B Proofs of propositions and derivatives

B.1 Proof Eq. (7) and (8)

Proof. To obtain derivative (7):

$$\begin{aligned}
\frac{\partial \tau^*(\lambda_m)}{\partial \overline{y}} &= \frac{1-\pi}{2\lambda_m}\left(\frac{\overline{y}(1-\pi)+\pi\underline{y}}{\lambda_m}\right)^{-1/2} \\
&= \frac{1-\pi}{2}\left(\overline{y}(1-\pi)+\pi\underline{y}\right)^{-1/2}\left(\frac{1}{\lambda_m}\right)^{-1/2}\frac{1}{\lambda_m} \\
&= \frac{1-\pi}{2}\left(\frac{1}{\overline{y}(1-\pi)+\pi\underline{y}}\right)^{1/2}\left(\frac{1}{\lambda_m}\right)^{1/2}
\end{aligned}$$

$$= \left(\frac{1-\pi}{2}\right)\left[\frac{1}{\lambda_m\left(\overline{y}(1-\pi)+\pi\underline{y}\right)}\right]^{1/2} > 0 \tag{B.1.1}$$

■

Proof of derivative (8) follows the same steps of the proof of derivative (7) and does not need to be proven again.

B.2 Proof Proposition 1

Proof. To derive derivative (10):

$$\begin{aligned}
\frac{\partial\tau^*(\lambda_m)}{\partial\pi} &= \left(\frac{\underline{y}-\overline{y}}{2\lambda_m}\right)\left(\frac{\overline{y}(1-\pi)+\pi\underline{y}}{\lambda_m}\right)^{-1/2} \\
&= -\left(\frac{\overline{y}-\underline{y}}{2}\right)\left(\frac{1}{\overline{y}(1-\pi)+\pi\underline{y}}\right)^{1/2}\left(\frac{1}{\lambda_m}\right)\left(\frac{1}{\lambda_m}\right)^{-1/2} \\
&= -\left(\frac{\overline{y}-\underline{y}}{2}\right)\left(\frac{1}{\overline{y}(1-\pi)+\pi\underline{y}}\right)^{1/2}\left(\frac{1}{\lambda_m}\right)^{1/2} \\
&= -\left[\frac{\left(\overline{y}-\underline{y}\right)^2}{4\lambda_m\left(\overline{y}(1-\pi)+\pi\underline{y}\right)}\right]^{1/2} < 0
\end{aligned} \tag{B.2.1}$$

■

B.3 Proof Proposition 2

Proof. To derive derivative (11):

$$\begin{aligned}
\frac{\partial^2\tau^*(\lambda_m)}{\partial\pi\partial\lambda_m} &= \left[\frac{\left(\overline{y}-\underline{y}\right)^2}{4\left(\overline{y}(1-\pi)+\pi\underline{y}\right)}\right]^{1/2}\frac{1}{2}\left(\frac{1}{\lambda_m^2}\right)\left(\frac{1}{\lambda_m}\right)^{-1/2} \\
&= \left[\frac{\left(\overline{y}-\underline{y}\right)^2}{16\left(\overline{y}(1-\pi)+\pi\underline{y}\right)}\right]^{1/2}\left(\frac{1}{\lambda_m}\right)^{3/2} > 0
\end{aligned} \tag{B.3.1}$$

■

C Social optimum

Proof. Noticing that, as the population is normalised to 1, $\int_{\underline{\lambda}}^{\overline{\lambda}} \kappa \mathrm{d}F(\lambda_i) = \kappa$, where κ is a constant. Thus, the welfare function can be rewritten as:

$$\begin{aligned} W(\tau) &= \pi\left(\frac{\underline{y}}{p(\tau)}\right) + (1-\pi)\left(\frac{\overline{y}}{p(\tau)}\right) + g(\tau) - E(\tau) - \tau\int_{\underline{\lambda}}^{\overline{\lambda}} \lambda_i dF(\lambda_i) \\ &= \pi\left(\frac{\underline{y}}{p(\tau)}\right) + (1-\pi)\left(\frac{\overline{y}}{p(\tau)}\right) + g(\tau) - E(\tau) - \tau\lambda_a \end{aligned} \tag{C.0.1}$$

It is straightforward that the solution to the government maximisation problem (12) is the same as the solution of voters' maximisation problem (5), where the average level of ideology, λ_a, replaces the individual level of ideology, λ_i. Notice also that $\widetilde{\lambda}$ remains the same as for the median voter problem. ■

D Economic inequality and social insurance policy

In this extension, we study how economic inequality and redistributive social insurance policies affect preferences for environmental policies. Henceforth, we adopt a simple definition of income inequality, that is the difference between incomes of low-income and high-income voters, denoted as $\Delta = \overline{y} - \underline{y}$. We also assume that the government already has a ranking-preserving redistributive policy, such that high-income voters pay a lump sum income tax $t > 0$ which is redistributed to low-income voters in the form of a lump-sum transfer, b, such that $\overline{y} - t > \underline{y} + b$. For simplicity, we assume that the government budget constraints for the environmental policy and the redistributive policy are perfectly separable, such that the tax revenue collected for the redistribution cannot be spent for the environmental policy; the reverse also holds. Therefore, the government budget constraint for the redistributive policy can be written as:

$$b = t\left(\frac{1-\pi}{\pi}\right) \tag{D.0.1}$$

As a consequence, we can write the post-tax income inequality as a function of t and π, such that the post-tax income inequality $\widetilde{\Delta}(t,\pi) = \overline{y} - t - \underline{y} - t(1-\pi)/\pi$, while the pre-tax income inequality remains $\Delta = \overline{y} - \underline{y}$. Instead, the indirect utility function can be rewritten as:

$$\mathbb{E}[V_i(\tau)] = \pi \left(\frac{\underline{y} + t(1-\pi)/\pi}{p(\tau)} \right) + (1-\pi) \left(\frac{\overline{y} - t}{p(\tau)} \right) + g(\tau) - \alpha Q(\tau) - \lambda_i \tau, \tag{D.0.2}$$

whereby $Q(\tau) = \pi(\underline{y} + t(1-\pi)/\pi) + (\overline{y} - t)(1-\pi) = \underline{y}\pi + \overline{y}(1-\pi)$, as income is completely redistributed from high-income voters to low-income voters. Rearranging the terms in the expected utility from consumption, the maximisation problem can be written as in Eq. (A.2.1) and it becomes clear the preferred tax function is not a function of the social insurance policy, t:

$$\tau^*(\lambda_i) = \begin{cases} \left(\frac{\overline{y} - \pi\Delta}{\lambda_i} \right)^{1/2} - \frac{1}{\alpha}, & \lambda_i < \widetilde{\lambda} \\ \\ 0, & \lambda_i \geq \widetilde{\lambda} \end{cases} \tag{D.0.3}$$

Thus, more generous social insurance systems do not change the support for redistribution or the negative effect of higher economic insecurity, such that:

$$\frac{\partial \tau^*(\lambda_m)}{\partial t} = 0, \qquad \frac{\partial^2 \tau^*(\lambda_m)}{\partial \pi \partial t} = 0 \tag{D.0.4}$$

A possible interpretation for results in Eqs. D.0.4 is that voters the marginal increase in the tax burden if high-income (due to higher t) cancels out with the marginal decrease in the tax burden (due to higher b) if low-income. As a consequence, the redistributive policy does not affect preferences for the environmental tax under income uncertainty. On the other hand, by taking the derivative of equation (6) with respect to Δ, we determine the direct effect of market inequality on support for climate policies:

$$\frac{\partial \tau^*(\lambda_m)}{\partial \Delta} = - \left[\frac{\pi^2}{4\lambda_m (\overline{y} - \pi\Delta)} \right]^{1/2} < 0 \tag{D.0.5}$$

Proof. See proof (D.1.1) ■

Derivative (D.0.5) shows that if market inequality increases, the support for climate policies decreases. Notice that derivative (D.0.5) equals the effect of a decline in the income of low-income voters, that is $\partial\tau^*(\lambda_m)/\partial\Delta = -\partial\tau^*(\lambda_m)/\partial\underline{y}$. Indeed, when market inequality is higher, voters perceive that becoming a low-income person would mean facing a larger consumption-tax burden on their income compared to high-income people. As a response, they reduce their support for environmental tax as insurance against possible economic adversities. Finally, by taking the cross-partial derivative of (D.0.5) with respect to the probability of becoming a low-income person, π, we observe that economic insecurity has a larger negative effect when market income inequality is higher:

$$\frac{\partial^2\tau^*(\lambda_m)}{\partial\pi\partial\Delta} = -\left(\frac{1}{2\lambda_m^{1/2}}\right)\left[(\overline{y}-\pi\Delta)^{-1/2} + \frac{\pi\Delta}{2}(\overline{y}-\pi\Delta)^{-3/2}\right] < 0 \tag{D.0.6}$$

Proof. See proof (D.1.3) ■

D.1 Proofs: Economic inequality

Proof. Proof of derivative (D.0.5):

$$\begin{aligned}\frac{\partial\tau^*(\lambda_m)}{\partial\Delta} &= -\frac{1}{2}\left(\frac{\overline{y}-\pi\Delta}{\lambda_m}\right)^{-1/2}\frac{\pi}{\lambda_m} \\ &= -\frac{\pi}{2}(\overline{y}-\pi\Delta)^{-1/2}\left(\frac{1}{\lambda_m}\right)^{1/2} \\ &= -\left[\frac{\pi^2}{4\lambda_m(\overline{y}-\pi\Delta)}\right]^{1/2} < 0 \end{aligned} \tag{D.1.1}$$

■

Proof. First, notice that:

$$\frac{\partial\tau^*(\lambda_m)}{\partial\pi} = -\frac{\Delta}{2\lambda_m}\left[\frac{\overline{y}-\pi\Delta}{\lambda_m}\right]^{-1/2}$$

$$= -\frac{\Delta\left(\overline{y} - \pi\Delta\right)^{-1/2}}{2\lambda_m^{1/2}} < 0 \tag{D.1.2}$$

Thus, proof of the cross-partial derivative of (D.0.6) is straightforward:

$$\frac{\partial^2 \tau^*(\lambda_m)}{\partial\pi\partial\Delta} = -\left(\frac{1}{2\lambda_m^{1/2}}\right)\left[\left(\overline{y} - \pi\Delta\right)^{-1/2} + \frac{\pi\Delta}{2}\left(\overline{y} - \pi\Delta\right)^{-3/2}\right] < 0 \tag{D.1.3}$$

■

E Government inefficiency

Differently from the benchmark model, we assume that the government is inefficient in providing the public good, $g(\tau)$. The inefficiency is captured by a parameter $\gamma \in [0, 1)$ such that for every unit of public good provided, $g(\tau)$, a portion of the good $\gamma g(\tau)$ is wasted. The inefficiency γ can be due to corruption or bad management of public resources. One can also interpret γ as a political mistrust parameter, which is often associated with higher inefficiency in the distribution of public resources, or as the uncertainty of the allocation of resources the environmental tax.

The difference $m^b \equiv \alpha - \gamma$ represents the net marginal benefit of implementing a green policy, which is decreasing in the government's inefficiency, γ. We assume that the net marginal benefit is such that $m^b = 0$ if $\alpha \leq \gamma$. In other words, if the marginal waste of resources while implementing the public policy is higher than or equal to the marginal damage people receive from pollution, voters will have no net marginal benefit from implementing the environmental policy. The indirect utility function becomes:

$$\mathbb{E}[V_i(\tau)] = \pi\left(\frac{\underline{y}}{p(\tau)}\right) + (1-\pi)\left(\frac{\overline{y}}{p(\tau)}\right) + (1-\gamma)g(\tau) - \alpha Q(\tau) - \lambda_i\tau \tag{E.0.1}$$

An interior solution can exist only if $\alpha > \gamma$ (See appendix E.1). In other words, an interior solution exists only if the marginal damage of an extra unit of pollution, α, is greater than the marginal waste of resources, γ. Instead, if the marginal waste of resources while

implementing the public policy is smaller than or equal to the marginal damage people receive from pollution (i.e. $\alpha \leq \gamma$), voters always choose $\tau^*(\lambda_i) = 0$.

Maximising the utility function (E.0.1) with respect to τ (See appendix E.2), we obtain the following preferred environmental tax function:

$$\tau^*(\lambda_i) = \begin{cases} \left[\frac{(\alpha-\gamma)\left(\overline{y}(1-\pi)+\pi\underline{y}\right)}{\alpha\lambda_i}\right]^{1/2} - \frac{1}{\alpha}, & \lambda_i < \widetilde{\lambda} \\ \\ 0, & \lambda_i \geq \widetilde{\lambda}, \end{cases} \tag{E.0.2}$$

whereby $\widetilde{\lambda} = \alpha(\alpha-\gamma)(\overline{y}(1-\pi)+\pi\underline{y})$ is the ideological no-policy threshold. Moreover, notice that if the marginal waste of resources while implementing the public policy is higher than or equal to the marginal damage that people receive from pollution (i.e., $\alpha \leq \gamma$), voters will always choose $\tau^*(\lambda_i) = 0$.

Assuming that $\alpha > \gamma$ and taking the derivative of the tax function (E.0.2) with respect to γ, we can show that higher government inefficiency reduces the support for the environmental tax as people expect the government will not turn all the tax revenue from τ into a valuable public green good:

$$\frac{\partial\tau^*(\lambda_m)}{\partial\gamma} = -\left[\frac{\left(\overline{y}(1-\pi)+\pi\underline{y}\right)}{4\left(\alpha-\gamma\right)\alpha\lambda_m}\right]^{1/2} < 0 \tag{E.0.3}$$

Proof. See proof (E.3.1) ■

Finally by taking the first-order derivative of the tax function (E.0.2) with respect to π and then the second-order cross-partial derivative with respect to γ, we show that the negative effect of economic insecurity on environmental policy support is large when the government is more inefficient:

$$\frac{\partial\tau^*(\lambda_m)}{\partial\pi} = -\left(\frac{\overline{y}-\underline{y}}{2}\right)\left[\frac{(\alpha-\gamma)}{\left(\overline{y}(1-\pi)+\pi\underline{y}\right)\alpha\lambda_m}\right]^{-1/2} < 0 \tag{E.0.4}$$

$$\frac{\partial^2\tau^*(\lambda_m)}{\partial\pi\partial\gamma} = -\left(\frac{\overline{y}-\underline{y}}{4}\right)\left[\frac{\left(\overline{y}(1-\pi)+\pi\underline{y}\right)\alpha\lambda_m}{(\alpha-\gamma)^3}\right]^{1/2} < 0 \tag{E.0.5}$$

Proof. See proofs (E.3.2) and (E.3.3). ■

E.1 Government inefficiency: Proof of single-peaked preferences

To prove that preferences are single-peaked in τ, we need to verify that $\frac{\partial^2 \mathbb{E}[V(\tau)]}{\partial^2 \tau} < 0$. Consider the indirect utility function:

$$\mathbb{E}[V_i(\tau)] = \frac{\overline{y}(1-\pi)+\pi\underline{y}}{1+\alpha\tau} + \tau\alpha\frac{\overline{y}(1-\pi)+\pi\underline{y}}{1+\alpha\tau} - \alpha\frac{\overline{y}(1-\pi)+\pi\underline{y}}{1+\alpha\tau} - \tau\lambda_i \tag{E.1.1}$$

The FOC are:

$$\frac{\partial \mathbb{E}[V_i(\tau)]}{\partial \tau} : \; -\frac{\gamma\alpha(\overline{y}(1-\pi)+\pi\underline{y})}{(1+\alpha\tau)^2} + \frac{\alpha^2(\overline{y}(1-\pi)+\pi\underline{y})}{(1+\alpha\tau)^2} - \lambda_i \leq 0 \tag{E.1.2}$$

The SOC are:

$$\frac{\partial^2 \mathbb{E}[V_i(\tau)]}{\partial^2 \tau} : \; \frac{2\alpha^2(\overline{y}(1-\pi)+\pi\underline{y})(\gamma-\alpha)}{(1+\alpha\tau)^3} < 0 \tag{E.1.3}$$

The SOC (E.1.3) the function $\mathbb{E}[V_i(\tau)]$ is strictly concave in τ if and only if the marginal damage of an extra unit of pollution, α, is greater than the marginal waste of resources, γ, such that $\alpha > \gamma$. Because by implementing the green policy, α represents a marginal decrease in the pollution damage, the difference $m^b \equiv \alpha - \gamma$ represents the net marginal benefit of implementing a green policy when a government is inefficient. Henceforth, we set a lower bound to the net marginal benefit, that is $m^b = 0$ if $\alpha - \gamma \leq 0$, such that voters cannot have a negative net marginal benefit if the government inefficiency is large enough compared to the per-unit damage of pollution.

E.2 Government inefficiency: maximisation problem

Agents solve the following maximisation problem:

$$\max_{\tau \geq 0} \quad \frac{\overline{y}(1-\pi)+\pi\underline{y}}{1+\alpha\tau} + (1-\gamma)\tau\alpha\frac{\overline{y}(1-\pi)+\pi\underline{y}}{1+\alpha\tau} - \alpha\frac{(\overline{y}(1-\pi)+\pi\underline{y})}{1+\alpha\tau} - \tau\lambda_i \tag{E.2.1}$$

To solve the maximisation problem, we apply the Lagrangian method. The Lagrangian function is:

$$\mathcal{L} = \frac{\overline{y}(1-\pi)+\pi\underline{y}}{1+\alpha\tau} + (1-\gamma)\tau\frac{\overline{y}(1-\pi)+\pi\underline{y}}{1+\alpha\tau} - \alpha\frac{(\overline{y}(1-\pi)+\pi\underline{y})}{1+\alpha\tau} - \tau\lambda_i + \nu_1\tau \tag{E.2.2}$$

where $\nu_1 \geq 0$ is the lagrangian multiplier.

The necessary conditions for a local maximum are given by the FOCs. The FOCs must solve the Kuhn-Tucker conditions: (A.2.3) -(A.2.5):

$$\frac{\partial\mathcal{L}}{\partial\tau} \leq 0: \quad -\frac{\gamma\alpha(\overline{y}(1-\pi)+\pi\underline{y})}{(1+\alpha\tau)^2} + \frac{\alpha^2(\overline{y}(1-\pi)+\pi\underline{y})}{(1+\alpha\tau)^2} - \lambda_i + \nu_1 \leq 0, \tag{E.2.3}$$

$$\nu_1\frac{\partial\mathcal{L}}{\partial\nu_1} = 0: \ \nu_1\tau = 0, \tag{E.2.4}$$

$$\nu_1 \geq 0 \tag{E.2.5}$$

Henceforth, we verify the existence of both an interior and a corner solution:

- **Case 1 (corner solution):** $\nu_1 > 0$ and $\tau = 0$.

The Kuhn-Tucker condition (E.2.3), then becomes:

$$\alpha(\alpha-\gamma)(\overline{y}(1-\pi)+\pi\underline{y}) - \lambda_i + \nu_1 \leq 0 \tag{E.2.6}$$

Because $\nu_1 > 0$ (E.2.6) is satisfied only if:

$$\lambda_i \geq \alpha(\alpha - \gamma)(\overline{y}(1-\pi) + \pi\underline{y}) \tag{E.2.7}$$

- **Case 2 (interior solution):** $\nu_1 = 0$ and $\tau > 0$.

An interior solution is found by setting (E.2.3) equal to 0 and solving it for τ:

$$-\frac{\gamma\alpha(\overline{y}(1-\pi) + \pi\underline{y})}{(1+\alpha\tau)^2} + \frac{\alpha^2(\overline{y}(1-\pi) + \pi\underline{y})}{(1+\alpha\tau)^2} = \lambda_i \tag{E.2.8}$$

which gives the following optimal tax rate $\tau^* > 0$ if $\alpha > \gamma$:

$$\tau^* = \left[\frac{(\alpha-\gamma)\left(\overline{y}(1-\pi) + \pi\underline{y}\right)}{\alpha\lambda_i}\right]^{1/2} - \frac{1}{\alpha} \tag{E.2.9}$$

E.3 Government inefficiency: proofs derivatives

Proof. Proof of derivative (E.3.1):

$$\begin{aligned}
\frac{\partial\tau^*(\lambda_m)}{\partial\gamma} &= -\frac{1}{2}\left[\frac{(\alpha-\gamma)\left(\overline{y}(1-\pi) + \pi\underline{y}\right)}{\alpha\lambda_m}\right]^{-1/2}\left[\frac{\left(\overline{y}(1-\pi) + \pi\underline{y}\right)}{\alpha\lambda_m}\right] \\
&= -\frac{1}{2}\left[\frac{\left(\overline{y}(1-\pi) + \pi\underline{y}\right)}{\alpha\lambda_m}\right]^{1/2}(\alpha-\gamma)^{-1/2} \\
&= -\left[\frac{\left(\overline{y}(1-\pi) + \pi\underline{y}\right)}{4(\alpha-\gamma)\alpha\lambda_m}\right]^{1/2} < 0
\end{aligned} \tag{E.3.1}$$

■

Proof. Proof of derivative (E.3.2):

$$\begin{aligned}
\frac{\partial\tau^*(\lambda_m)}{\partial\pi} &= \frac{1}{2}\left[\frac{(\alpha-\gamma)\left(\overline{y}(1-\pi) + \pi\underline{y}\right)}{\alpha\lambda_m}\right]^{-1/2}\left(\frac{\underline{y}-\overline{y}}{\alpha\lambda_m}\right) \\
&= -\left(\frac{\overline{y}-\underline{y}}{2}\right)\left[(\alpha-\gamma)\left(\overline{y}(1-\pi) + \pi\underline{y}\right)\alpha\lambda_m\right]^{-1/2} < 0
\end{aligned} \tag{E.3.2}$$

■

Proof. Proof of the cross-partial derivative of (E.3.3):

$$\frac{\partial^2 \tau^*(\lambda_m)}{\partial \pi \partial \gamma} = -\left(\frac{\overline{y}-\underline{y}}{4}\right)\left[\frac{\alpha-\gamma}{\left(\overline{y}(1-\pi)+\pi\underline{y}\right)\alpha\lambda_m}\right]^{-3/2}\left[\frac{1}{\left(\overline{y}(1-\pi)+\pi\underline{y}\right)\alpha\lambda_m}\right]$$

$$= -\left(\frac{\overline{y}-\underline{y}}{4}\right)\left[\frac{1}{\left(\overline{y}(1-\pi)+\pi\underline{y}\right)\alpha\lambda_m}\right]^{-1/2}(\alpha-\gamma)^{-3/2}$$

$$= -\left(\frac{\overline{y}-\underline{y}}{4}\right)\left[\frac{\left(\overline{y}(1-\pi)+\pi\underline{y}\right)\alpha\lambda_m}{(\alpha-\gamma)^3}\right]^{1/2} < 0 \tag{E.3.3}$$

■

F PCA analysis

F.1 PCA for preferences for climate policies

Table F.1: Principal components/correlation

	(1) Eigenvalue	(2) Difference	(3) Proportion	(4) Cumulative
Comp1	1.233226	.4664521	0.6166	0.6166
Comp2	.7667739	.	0.3834	1.0000

Notes: the table presents the principal components of the PCA analysis. Columns report information about Eigenvalue (1); Difference (2); Proportion (3); and Cumulative (4).

Table F.2: Principal components (eigenvectors)

	(1) Comp1	(2) Comp2	(3) Unexplained
Increasing taxes on fossil fuels	.7071068	.7071068	0
Subsidise renewable energy	.7071068	-.7071068	0

Notes: the graph shows the weight of each variable for each component of the PCA analysis.

Table F.3: Pairwise correlations

	Increasing taxes on fossil fuels	Subsidise renewable energy
Increasing taxes on fossil fuels	1	
Subsidise renewable energy	0.233***	1

Notes: the table shows the pairwise correlation for the variables used in the PCA analysis. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

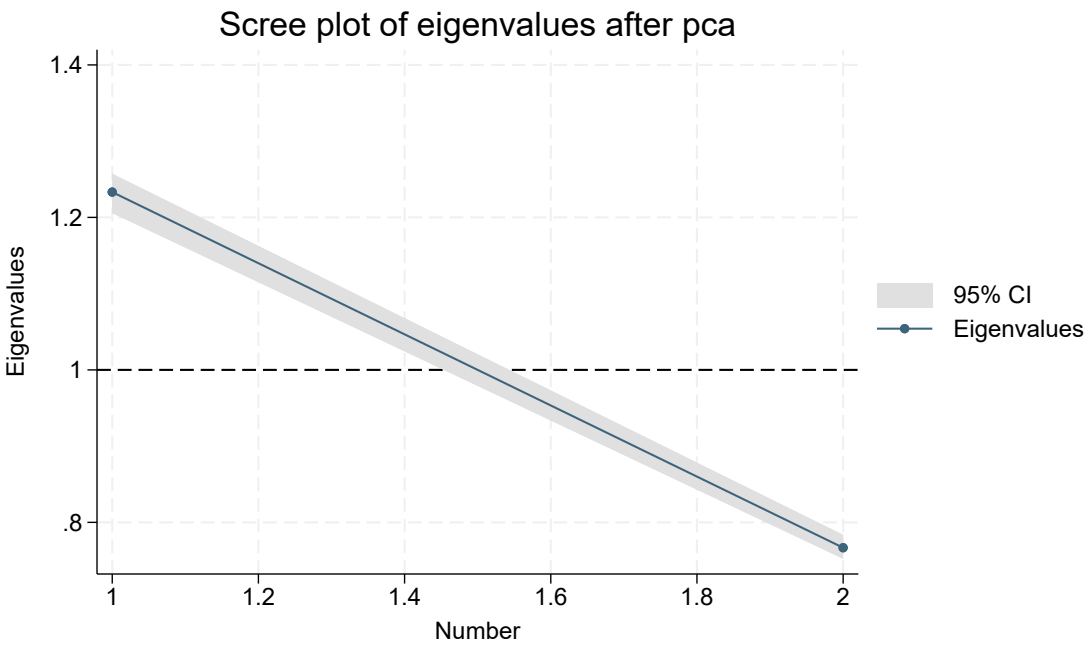

Figure F.1: Screen plot of eigenvalues after PCA

Notes: the graph shows the proportion of variance explained by each component with 95% confidence intervals.

F.2 PCA for economic insecurity

Table F.4: Principal components/correlation

	(1) Eigenvalue	(2) Difference	(3) Proportion	(4) Cumulative
Comp1	1.677665	.8071779	0.5592	0.5592
Comp2	.8704874	.4186401	0.2902	0.8494
Comp3	.4518473	.	0.1506	1.0000

Notes: the table presents the principal components of the PCA analysis. Columns report information about Eigenvalue (1); Difference (2); Proportion (3); and Cumulative (4).

Table F.5: Principal components (eigenvectors)

	(1) Comp1	(2) Comp2	(3) Comp3	(4) Unexplained
Ever unemployed for more than three months	.4005635	.916269	.0001062	0
Not enough money for the next 12 months	.6478885	-.2833181	.7070865	0
Feeling about household's income nowadays	.6479115	-.2831642	-.7071271	0

Notes: the graph shows the weight of each variable for each component of the PCA analysis.

Table F.6: Pairwise correlations

	Ever unemployed	Not enough money	Feeling income nowadays
Ever unemployed for more than three months	1		
Not enough money for the next 12 months	0.209***	1	
Feeling about household's income nowadays	0.210***	0.548***	1

Notes: the table shows the pairwise correlations for the variables used in the PCA analysis. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

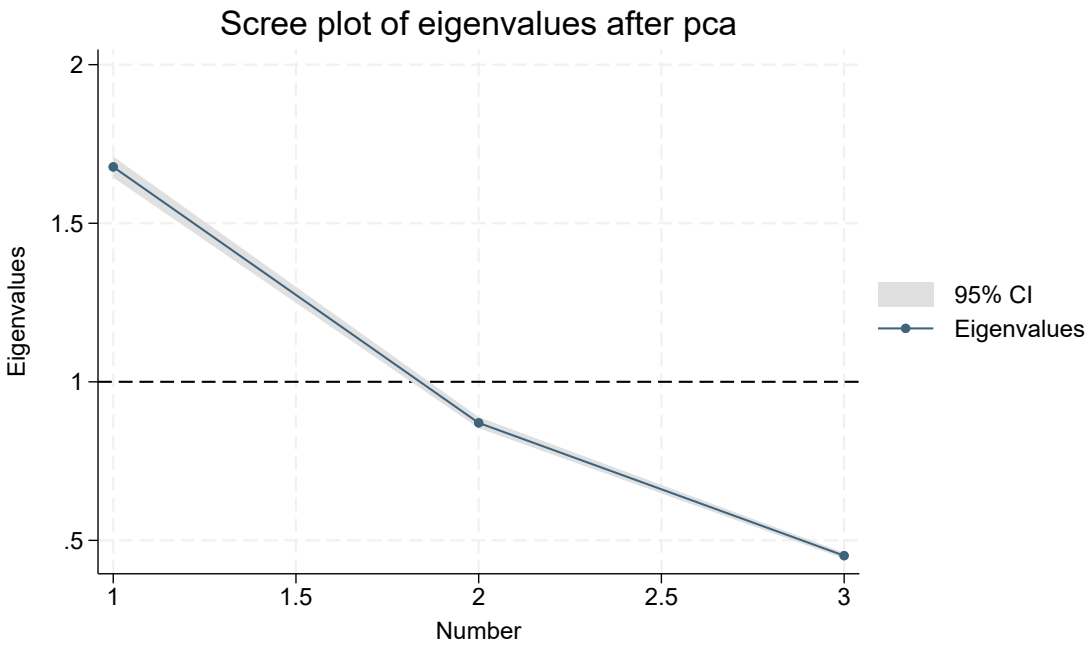

Figure F.2: Screen plot of eigenvalues after PCA

Notes: the graph shows the proportion of variance explained by each component with 95% confidence intervals.

G Regional aggregation

Some regions have been aggregated to merge the ESS dataset with the QoG EU regional datase. The aggregation has been necessary for two reasons: (i) the hierarchical level in the ESS and the EU regional dataset did not always coincide (e.g., some regions were NUTS2 in one dataset and NUTS3 in the other dataset). In this case, the regions of the lower hierarchical level have been aggregated to the next higher hierarchical level. (ii) The definition of NUTS changed between the years 2016-2017 of the ESS and the 2020 version of the EU regional dataset. We aggregate regions following the instructions from Eurostat. Here below, we report the comprehensive list of regions used in the final analysis. We split the dataset into "regions that did not change", that is regions that matched in the ESS and the QoG EU regional dataset and "regions that changed", that is regions that have been aggregated to match the two datasets.

G.1 List of regions that did not change

- **Belgium** (NUTS 2): Antwerp (BE21), Limburg (BE22), East Flanders (BE23), Flemish Brabant (BE24), West Flanders (BE25), Hainaut (BE32), Liège (BE33), Luxembourg (BE34), Namur (BE35).

- **France** (NUTS 2): Île de France (FR10), Centre-Val de Loire (FRB0), Bourgogne-Franche-Comté (FRC1), Franche-Comté (FRC2), Lower Normandy (FRD1), Upper Normandy (FRD2), Hauts-de-France (FRE1), Picardy (FRE2), Alsace (FRF1), Champagne-Ardenne (FRF2), Lorraine (FRF3), Pays de la Loire (FRG0), Brittany (FRH0), Aquitaine (FRI1), Limousin (FRI2), Poitou-Charentes (FRI3), Languedoc-Roussillon (FRJ1), Midi-Pyrénées (FRJ2), Auvergne (FRK1), Rhône-Alpes (FRK2), Provence-Alpes-Côte d'Azur (FRL0).

- **Germany** (NUTS 1): Baden-Württemberg (DE1), Bavaria (DE2), Brandenburg (DE4), Hesse (DE7), Mecklenburg-Vorpommern (DE8), Lower Saxony (DE9), North

Rhine-Westphalia (DEA), Rhineland-Palatinate (DEB), Saxony (DED), Saxony-Anhalt (DEE), Schleswig-Holstein (DEF), Thuringia (DEG).

- **Italy** (NUTS 2): Piemonte (ITC1), Valle d'Aosta (ITC2), Liguria (ITC3), Lombardia (ITC4), Abruzzo (ITF1), Campania (ITF3), Puglia (ITF4), Basilicata (ITF5), Calabria (ITF6), Sicilia (ITG1), Sardegna (ITG2), Trentino-Alto Adige/Südtirol (ITH1), Friuli-Venezia Giulia (ITH2), Veneto (ITH3), Emilia-Romagna (ITH4), Toscana (ITH5), Umbria (ITI1), Marche (ITI2), Lazio (ITI3), Molise (ITI4).

- **Netherlands** (NUTS 2): Groningen (NL11), Friesland (NL12), Drenthe (NL13), Overijssel (NL21), Gelderland (NL22), Flevoland (NL23), Utrecht (NL31), Noord-Holland (NL32), Zuid-Holland (NL33), Zeeland (NL34), Noord-Brabant (NL41), Limburg (NL42).

- **Norway** (NUTS 2): Hedmark og Oppland (or Innlandet) (NO02), Trøndelag (NO06), Nord-Norge (NO07).

- **Portugal** (NUTS 2): Norte (PT11), Algarve (PT15), Centro (PT16), Península de Setúbal (PT17), Grande Lisboa (PT18).

- **Spain** (NUTS 2): A Coruña (ES11), Asturias (ES12), Cantabria (ES13), Álava/Araba (ES21), Navarre (ES22), La Rioja (ES23), Huesca (ES24), Madrid (ES30), Ávila (ES41), Castile-La Mancha (ES42), Extremadura (ES43), Barcelona (ES51), Alicante (ES52), Eivissa i Formentera (ES53), Almería (ES61), Murcia (ES62), El Hierro (ES70).

- **United Kingdom** (NUTS 1): North East (UKC), North West (UKD), Yorkshire and the Humber (UKE), East Midlands (UKF), West Midlands (UKG), East of England (UKH), London (UKI), South East (UKJ), South West (UKK), Wales (UKL), Scotland (UKM), Northern Ireland (UKN)

G.2 List of regions that changed

Below, we present the list of regions created by aggregating the ESS region to adapt to the QoG EU regional dataset. The changes below have been made following the NUTS changes reported on the Eurostat website (see `https://ec.europa.eu/eurostat/web/nuts/history`):

- **Ireland** (NUTS 2): Northern and Western Region (IE04), Southern Region (IE05), Eastern and Midland Region (IE06).
- **Czech Republic** (NUTS 2): Prague (CZ01), Central Bohemian Region (CZ02), South Bohemian Region (CZ03), Karlovy Vary Region (CZ04), Liberec Region (CZ05), Vysočina Region (CZ06), Olomouc Region (CZ07), Moravian-Silesian Region (CZ08).
- **Hungary** (NUTS 2): Pest (HU12), Central Transdanubia (HU21), Western Transdanubia (HU22), Southern Transdanubia (HU23), Northern Hungary (HU31), Northern Great Plain (HU32), Southern Great Plain (HU33).
- **Slovenia** (NUTS 2): Vzhodna Slovenija (SI03), Zahodna Slovenija (SI04).
- **Lithuania** (NUTS 2): Sostinės regionas (LT01), Vidurio ir vakaru Lietuvos regionas (LT02).
- **Poland** (NUTS 2): Malopolskie (PL21), Slaskie (PL22), Wielkopolskie (PL41), Zachodniopomorskie (PL42), Lubuskie (PL43), Dolnoslaskie (PL51), Kujawsko-Pomorskie (PL61), Warminsko-Mazurskie (PL62), Pomorskie (PL63), Lodzkie (PL71), Swietokrzyskie (PL72), Lubelskie (PL81), Podkarpackie (PL82), Podlaskie (PL84), Mazowiecki regionalny (PL92)
- **Finland** (NUTS 2): West Finland (FI19), Helsinki-Uusimaa (FI1B), South Finland (FI1C), North & East Finland (FI1D)

- **Sweden** (NUTS 2): Stockholm County (SE11), East Middle Sweden (SE12), Uppsala County (SE21), Södermanland County (SE22), Östergötland County (SE23), Örebro County (SE31), Västmanland County (SE32), South Sweden (SE33)

H Descriptive statistics

H.1 Control variable description

In our main econometric model, we control for the following variables:

- **Income controls:** income deciles, occupation (2-digit ISCO-08), employment (paid work, unemployed, education, disabled, retired, other)
- **Individual controls:** gender (a dummy taking value 1 if the respondent is a male), the highest level of completed education (ISCED), age, age squared, number of household members, type of respondent's domicile (big city, suburbs of a big city, town, country village or farm), marital status (married, divorced, widowed, none of these) and for citizenship (a dummy taking value 1 if the respondent is a citizen)
- **Ideology controls:** degree of religiosity (measured on a 0-10 scale), left-right scale (measured on a 0-10 scale), trust in politicians (measured on a 0-10 scale) and climate change attitudes (measured on a 0-100 scale))
- **Regional controls:** Population density per km^2, log of GDP per capita, percentage of the population with tertiary education, long-term unemployment as a percentage of total unemployment, blue-white collar ratio and total fertility rate

H.2 Climate change attitudes index

To take into account the different dimensions of environmental ideology, we build the PCA index using six ESS questions: (i) "*To what extent do you feel a personal responsibility to try*

to reduce climate change?", taking values going from 0 ("*Not at all*") to 10 ("*A great deal*"); (ii) "*How worried are you about climate change?*", taking values going from 1 ("*Not at all worried*") to 5 ("*Extremely worried*"); (iii) "*How much have you thought about climate change before today?*", taking values going from 1 ("*Not at all*") to 5 ("*A great deal*"); (iv) "*Do you think the world's climate is changing?*", taking values going from 1 ("*Definitely changing*") to 3 ("*Probably not changing*"); (v) "*How good or bad do you think the impact of climate change will be on people across the world?*", taking values going from 0 ("*Extremely good*") to 10 ("*Extremely bad*"); (vi) "*Do you think that climate change is caused by natural processes, human activity, or both?*", taking values going from 1 ("*Entirely by natural processes*") to 5 ("*Entirely by human activity*").[2] We reversed the values of questions (iv) and (v) so that higher values correspond to higher environmental concerns. The first principal component explains 42.67% of the total variance, and the eigenvalue of the principal component equals 2.560. We normalised the index such that it takes values in the interval $[0, 100]$. The higher the PCA index is, the higher is the individual level of environmental concern. Detailed results of the PCA analysis are presented in the next Section H.2.1.

[2]The original question (vi) also has the answer "I don't think climate change is happening" to which a value of 55 is assigned. In the full sample, only 349 observations chose this answer (i.e., 0.82% of the full sample). Of these observations, 174 observations are from Israel and Russia, which are not included in our analysis. However, after cleaning the dataset, no observations with this answer remain in the sample. Therefore, the Likert scale for question (vi) ranges from 1 to 5.

H.2.1 PCA for attitudes towards climate change

Table H.1: Principal components/correlation

	(1) Eigenvalue	(2) Difference	(3) Proportion	(4) Cumulative
Comp1	2.560265	1.639311	0.4267	0.4267
Comp2	.9209547	.1583557	0.1535	0.5802
Comp3	.762599	.0451847	0.1271	0.7073
Comp4	.7174142	.1446565	0.1196	0.8269
Comp5	.5727577	.1067487	0.0955	0.9223
Comp6	.466009	.	0.0777	1.0000

Notes: the table presents the principal components of the PCA analysis. Columns report information about Eigenvalue (1); Difference (2); Proportion (3); and Cumulative (4).

Table H.2: Principal components (eigenvectors)

	(1) Comp1	(2) Comp2	(3) Comp3	(4) Comp4	(5) Comp5	(6) Comp6	(7) Unexplained
Feeling responsible	.3938389	-.5566299	.2828344	-.0965148	.6281952	.226085	0
Being worried	.4959967	-.1719988	-.0482728	-.0894659	-.1565149	-.830405	0
Thinking about climate change	.4514063	-.2872854	-.2676877	-.1764331	-.6168839	.4799671	0
Climate is changing	.3846293	.2278403	-.4910431	.6965203	.2573342	.0875466	0
Climate change impact	.342068	.6250711	-.1542484	-.617405	.278806	.0977817	0
Natural processes vs human activity	.3607775	.3679967	.7623004	.2919291	-.237433	.1082546	0

Notes: the graph shows the weight of each variable for each component of the PCA analysis.

Table H.3: Pairwise correlations

	V_1	V_2	V_3	V_4	V_5	V_6
V_1: Feeling responsible	1					
V_2: Being worried	0.440***	1				
V_3: Thinking about climate change	0.386***	0.509***	1			
V_4: Climate is changing	0.219***	0.369***	0.325***	1		
V_5: Climate change impact	0.145***	0.318***	0.263***	0.262***	1	
V_6: Natural processes vs human activity	0.245***	0.332***	0.235***	0.262***	0.276***	1

Notes: the table shows the pairwise correlations for the variables used in the PCA analysis. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

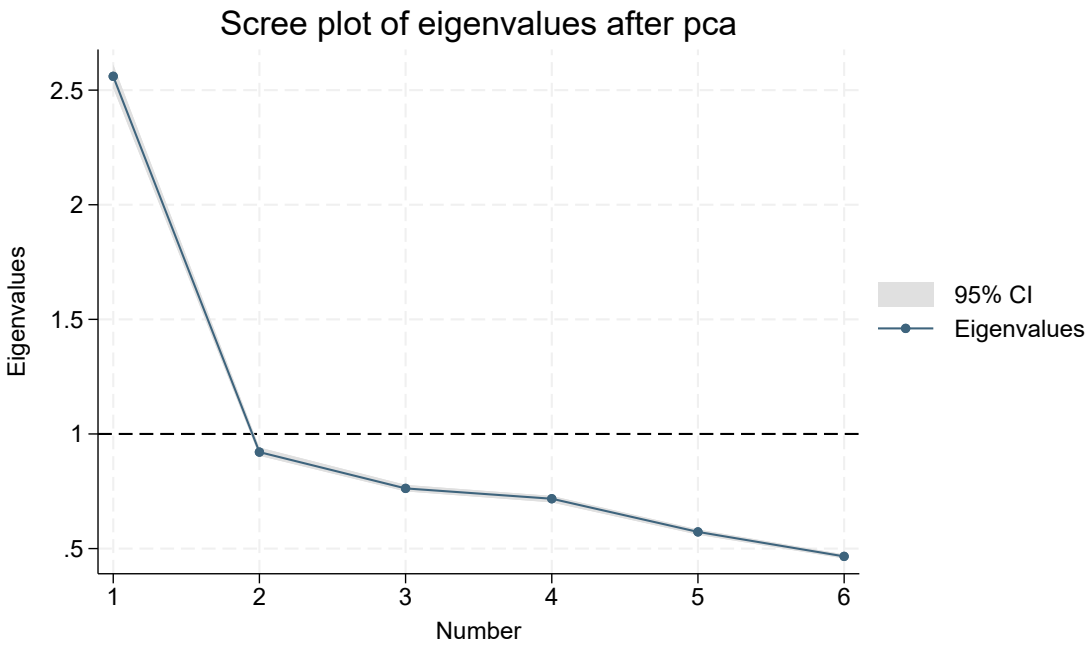

Figure H.1: Screen plot of eigenvalues after PCA

Notes: the graph shows the proportion of variance explained by each component with 95% confidence intervals.

H.3 Descriptive statistics tables

Table H.4: Descriptive statistics: regional-level data

Variables	(1) N	(2) Mean	(3) S.D.	(4) Min	(5) Max
Density	18,604	296.8	605.4	3.400	6,712
log(GDP)	18,604	10.86	0.762	8.486	13.38
% tertiary education 25-64	18,604	33.88	9.568	13.60	57.63
Long-term unemployment (% total unemployment)	18,604	38.64	11.06	12.80	68.10
Blue-white collar ratio	18,604	0.302	0.174	0.0413	0.854
Fertility rate, total	18,604	1.636	0.186	1.030	1.990

Notes: the table presents the descriptive statistics of the variables used in the analysis. The descriptive statistics are: total number of observations (1); mean (2); standard deviation (3); minimum value (4); maximum value (5).

Table H.5: Descriptive statistics: individual-level data

Variables	(1) N	(2) Mean	(3) S.D.	(4) Min	(5) Max
Economic insecurity	18,604	29.22	23.08	0	100
Climate change attitudes	18,604	58.31	16.74	0	100
Support for climate policies	18,604	62.26	22.17	0	100
Income	18,604	5.574	2.726	1	10
Left-Right scale	18,604	5.044	2.182	0	10
Trust in parliament	18,604	4.710	2.445	0	10
Paid work	18,604	0.570	0.495	0	1
Education	18,604	0.0370	0.189	0	1
Unemployed	18,604	0.0468	0.211	0	1
Permanently sick or disabled	18,604	0.0242	0.154	0	1
Retired	18,604	0.259	0.438	0	1
Housework	18,604	0.0541	0.226	0	1
Other	18,604	0.00887	0.0938	0	1
A big city	18,604	0.173	0.378	0	1
Suburbs or outskirts of big city	18,604	0.113	0.317	0	1
Town or small city	18,604	0.332	0.471	0	1
Country village	18,604	0.307	0.461	0	1
Farm or home in countryside	18,604	0.0743	0.262	0	1
Married	18,604	0.553	0.497	0	1
Separated	18,604	0.107	0.309	0	1
Widowed	18,604	0.0722	0.259	0	1
None of these	18,604	0.268	0.443	0	1
Less than lower secondary education (ISCED 0-1)	18,604	0.0816	0.274	0	1
Lower secondary education completed (ISCED 2)	18,604	0.137	0.344	0	1
Upper secondary education completed (ISCED 3)	18,604	0.360	0.480	0	1
Post-secondary non-tertiary education completed (ISCED 4)	18,604	0.150	0.357	0	1
Tertiary education completed (ISCED 5-6)	18,604	0.271	0.444	0	1
Religiosity	18,604	4.369	3.102	0	10
Isco (2-digits)	18,604	47.08	24.48	0	96
Household members	18,604	2.535	1.268	1	12
Age	18,604	50.92	16.94	15	98
Age (squared)	18,604	2,879	1,757	225	9,604
Male	18,604	0.500	0.500	0	1
Citizen	18,604	0.970	0.171	0	1
Ban energy inefficient sale	18,604	3.612	1.171	1	5
End of interview, year	18,604	2,016	0.475	2,016	2,017
Worried: energy too expensive (standardise)	18,568	0	1.000	-2.214	1.856
Support: ban energy inefficient sale (standardise)	18,604	0	1.000	-2.231	1.186
Support: biomass energy (standardise)	17,871	0	1.000	-2.138	1.404
Support: wind power (standardise)	18,444	0	1.000	-3.028	1.003
Support: solar power (standardise)	18,468	0	1.000	-3.432	0.947
Support: hydroelectric power (standardise)	18,363	0	1.000	-2.956	1.207

Notes: the table presents the descriptive statistics of the variables used in the analysis. The descriptive statistics are: total number of observations (1); mean (2); standard deviation (3); minimum value (4); maximum value (5).

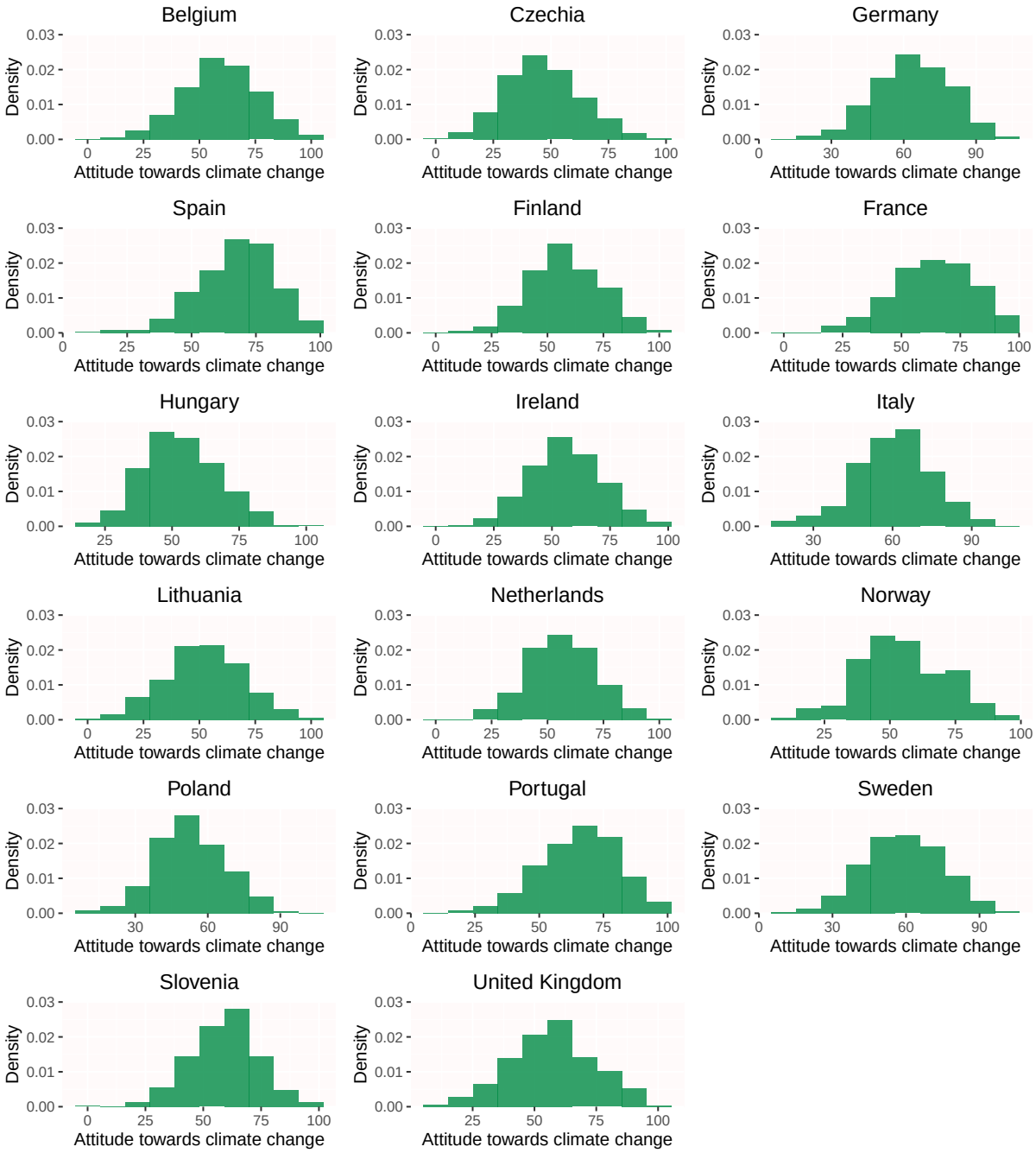

Figure H.2: Distribution of climate change attitudes by country

Notes: the figure shows the distribution of individual climate change attitudes for each country in our sample.

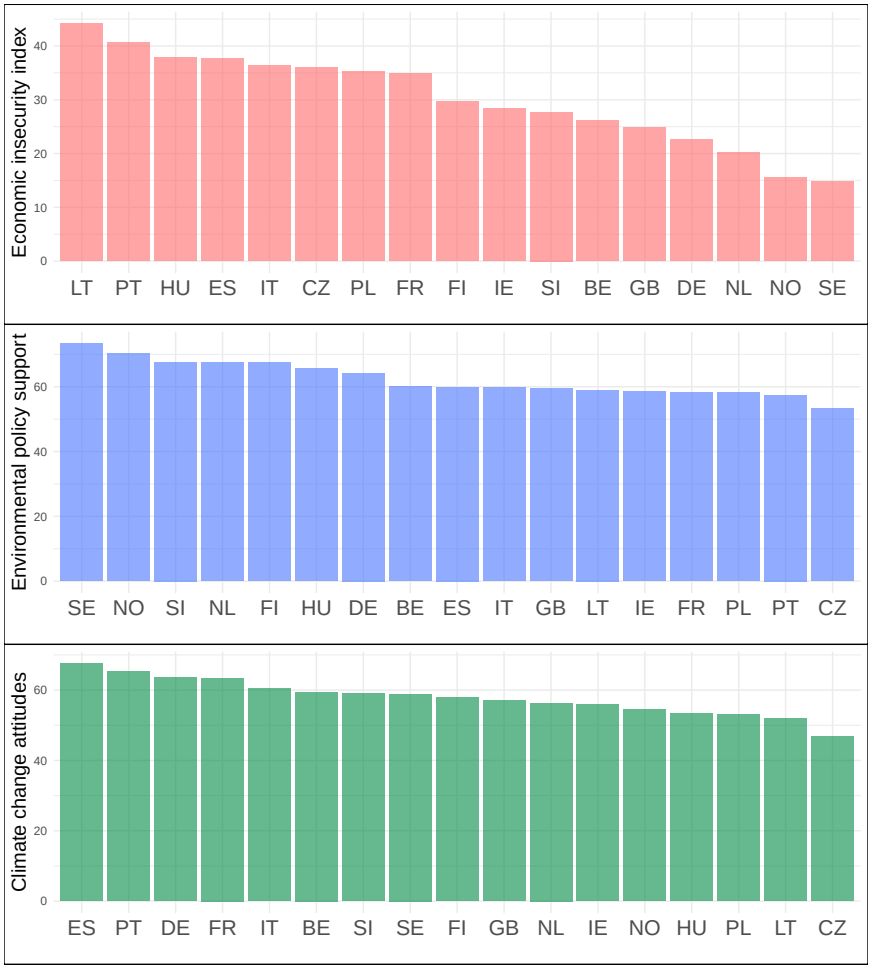

Figure H.3: Average level of perceived economic insecurity, support for environmental policies and climate change attitudes by country

Notes: the figure shows the bar charts for the average level of perceived economic insecurity, support for environmental policies and climate change attitudes for each country of our final individual data sample.

I Bivariate analysis

As a first descriptive evidence, in Figure I.1 we plot the scatterplot of the average environmental environmental policy support and economic insecurity across the 17 European countries in our sample. The scatterplot shows that, on average, countries with lower economic insecurity, such as Sweden and Norway, have higher support for climate policies. On the other hand, as economic insecurity increases, the average support for climate policies declines across countries.

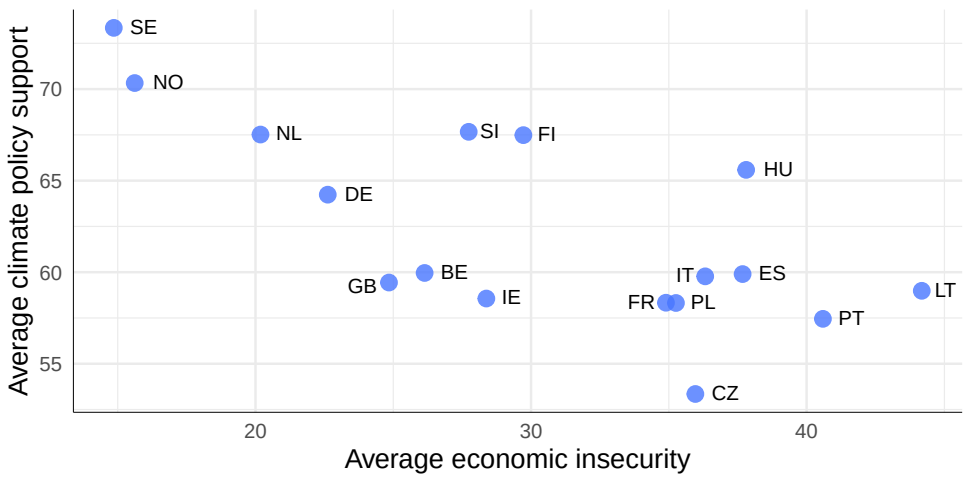

Figure I.1: Scatterplot of average economic insecurity on average support of environmental policies

Notes: the figure shows the scatterplot between the average economic insecurity on average support of environmental policies across the 17 European countries in our individual data sample (correlation: -0.445; p-value<0.01).

In Figure I.2, we focus on the distributions of the environmental policy support and the economic insecurity indices in each of the 17 European countries in our sample. A noteworthy pattern in Figure I.2 is a symmetry in the distributions of the two indices: the more the distribution of economic insecurity is skewed to the right — meaning that many people feel highly economically secure — the more the distribution of environmental policy support is skewed to the left — meaning that many people strongly support the introduction of environmental policies. For instance, countries with high economic security, such as Norway

or Sweden, are characterised by heavily "twin-peaked" overlapping distributions; on the other hand, countries with less skewed economic insecurity or environmental policy support distributions, such as Italy, Lithuania, and Spain or Portugal, do not show a "twin-peaks". As a consequence, Figure I.2 suggests a negative relationship between economic insecurity and support for environmental policy. However, as Figure I.2 and Figure I.1 are only for descriptive purposes, in the next subsection, we continue our analysis with a fixed-effects OLS regression model.

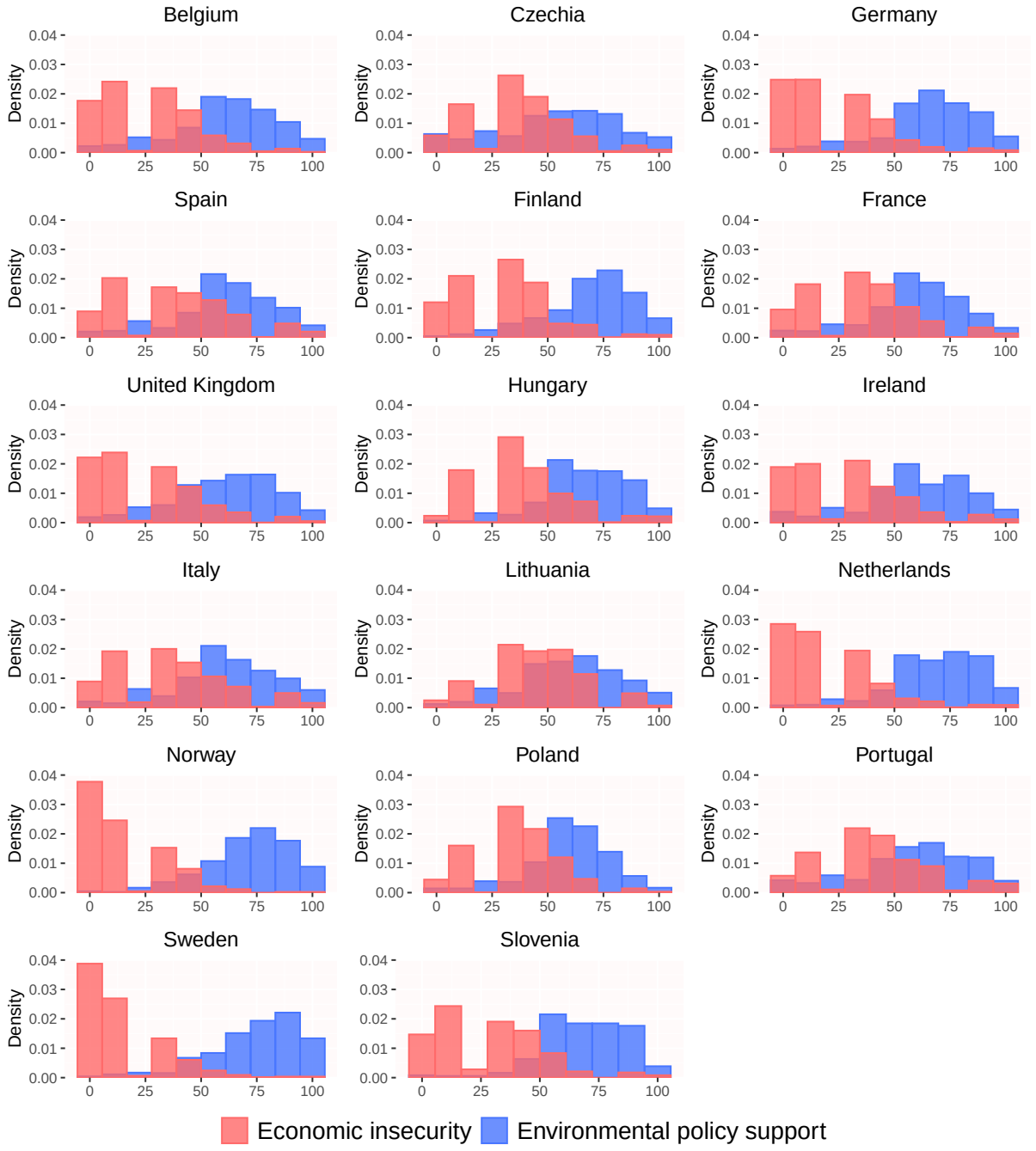

Figure I.2: Distribution of individual economic insecurity and support of environmental policies by country

Notes: the figure shows the distribution of individual economic insecurity (red histograms) and support of environmental policies (blue histograms) for each country in our sample.

J Risk attitudes and prudence variables

These variables come from the ESS questions, following the structure "*Now I will briefly describe some people. Please listen to each description and tell me how much each person is or is not like you*": (i) "*She/he looks for adventures and likes to take risks. She/he wants to have an exciting life.*", taking values going from 1 ("*Very much like me*") to 6 ("*Not like me at all*"); (ii) "*It is important to her/him to live in secure surroundings. She/he avoids anything that might endanger her/his safety.*", taking values going from 1 ("*Very much like me*") to 6 ("*Not like me at all*"). We create two dummy variables —namely, "Risk averse" and "Prudent"— assigning value 1 if respondents select answers between 1 ("*Very much like me*") and 3 ("*Somewhat like me*") and 0 otherwise.

K Robustness checks

Table K.1: Regression: heterogeneity of climate change attitudes with regional controls

	Environmental policies support			
Variables	(1)	(2)	(3)	(4)
Economic insecurity	-0.046***	-0.046***	-0.046***	-0.045***
	(0.012)	(0.011)	(0.011)	(0.011)
Climate change attitudes	0.424***	0.434***	0.338***	0.360***
	(0.022)	(0.021)	(0.022)	(0.020)
Climate change attitudes × High long-term unemployment	-0.065**			
	(0.030)			
Climate change attitudes × High blue-white collar employment ratio		-0.083***		
		(0.029)		
Climate change attitudes × High share of tertiary educated			0.103***	
			(0.030)	
Climate change attitudes × High GDP per capita				0.063**
				(0.028)
Regions	160	160	160	160
Observations	18,604	18,604	18,604	18,604
Adjusted R-squared	0.192	0.193	0.193	0.192

Notes: Columns (1)-(4) show the interaction effect between climate change attitudes and regions with high long-term unemployment (1), a high blue-white collar employment ratio (2), a high share of tertiary educated people (3) and a high GDP per capita (4). Aggregate regional controls, income controls, individual controls, ideology controls and country- and year-fixed effects are always included. Each variable interacting with climate change attitudes is included as a control in the regression but not shown in the table. Standard errors clustered at the regional level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

Table K.2: Regression: heterogeneity of climate 'pro-environmental' attitudes with regional unemployment

	Environmental policies support		
Variables	(1)	(2)	(3)
Economic insecurity	-0.046***	-0.041***	-0.041***
	(0.011)	(0.011)	(0.011)
Long-term unemployment	-0.133**	-0.136**	-0.097
	(0.062)	(0.062)	(0.066)
Climate change attitudes	0.391***		
	(0.015)		
Pro-environmental attitudes		9.869***	12.796***
		(0.351)	(1.135)
Pro-environmental attitudes × Long-term unemployment			-0.076***
			(0.028)
Regions	160	160	160
Observations	18,604	18,604	18,604
Adjusted R-squared	0.192	0.163	0.163

Notes: Column (1) is Column (5) in Table 1. Column (2) introduces the 'pro-environmental' dummy variable instead of the continuous "Climate change attitudes" control. Column (3) shows the interaction between regional unemployment and the 'pro-environmental' dummy. Aggregate regional controls, income controls, individual controls, ideology controls and country- and year-fixed effects are always included. Standard errors clustered at the regional level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

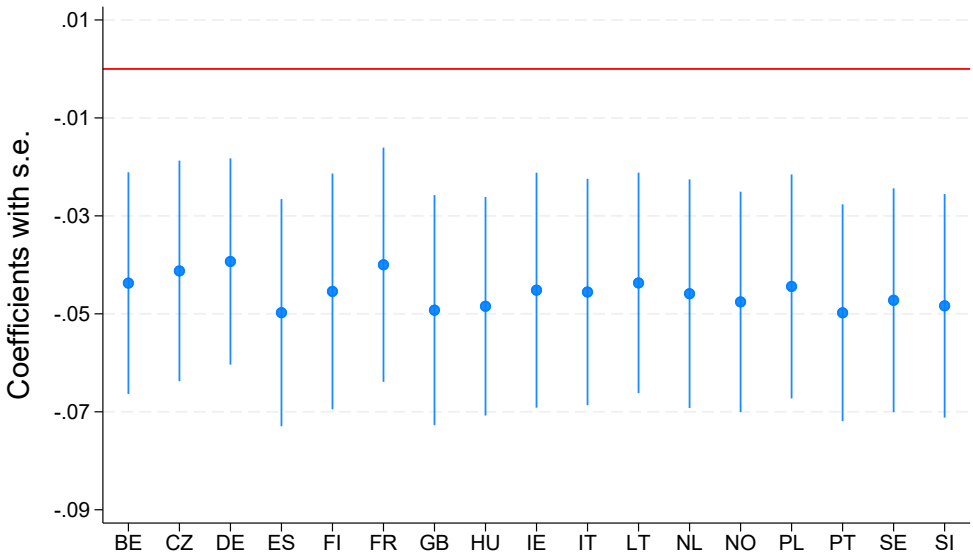

Figure K.1: Leave-one-country-out analysis

Notes: the graph shows the estimation of the econometric model (14) each time leaving a different country out of the sample. A country that is removed from one estimation is then reintroduced in the following estimations. The labels on the x-axis represent the ISO alpha-2 code of the country excluded in each estimation. Confidence intervals are at 95% level.

Table K.3: Regression: Support for environmental policies and perceived economic insecurity with nonlinear income

	Environmental policies support				
Variables	(1)	(2)	(3)	(4)	(5)
Economic insecurity	-0.088***	-0.081***	-0.041***	-0.041***	-0.046***
	(0.012)	(0.011)	(0.012)	(0.012)	(0.011)
Income Decile 2			-1.311*	-1.243	-1.816**
			(0.764)	(0.784)	(0.746)
Income Decile 3			1.356	1.501*	1.033
			(0.889)	(0.901)	(0.811)
Income Decile 4			0.262	0.201	-0.163
			(0.855)	(0.886)	(0.823)
Income Decile 5			0.457	0.446	0.027
			(1.046)	(1.086)	(0.999)
Income Decile 6			0.919	0.866	0.099
			(1.122)	(1.129)	(0.979)
Income Decile 7			1.133	1.076	0.201
			(1.030)	(1.072)	(0.926)
Income Decile 8			1.431	1.235	0.458
			(1.043)	(1.075)	(0.957)
Income Decile 9			2.144*	1.830	0.541
			(1.103)	(1.116)	(1.007)
Income Decile 10			2.699**	2.145*	0.935
			(1.107)	(1.088)	(0.985)
Climate change attitudes					0.391***
					(0.015)
Country-Year FE	✓	✓	✓	✓	✓
Regional Controls		✓	✓	✓	✓
Income Controls			✓	✓	✓
Individual Controls				✓	✓
Ideology Controls					✓
Regions	160	160	160	160	160
Observations	18,604	18,604	18,604	18,604	18,604
Adjusted R-squared	0.061	0.068	0.091	0.100	0.192

Notes: Columns (1)-(5) report the support for environmental policies regressed on perceived economic insecurity (1), aggregate regional controls (2), income controls (3), individual controls (4) and ideology controls (5). Income is introduced as a categorical variable (reference category: Income decile 1). Country- and year-fixed effects are always included. Standard errors clustered at the regional level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

Table K.4: Regression: income and perceived economic insecurity separately

	Environmental policies support				
Panel (a): income only	(1)	(2)	(3)	(4)	(5)
Income	0.909***	0.816***	0.450***	0.405***	0.300***
	(0.103)	(0.098)	(0.097)	(0.099)	(0.091)
Country-Year FE	✓	✓	✓	✓	✓
Regional Controls		✓	✓	✓	✓
Income Controls			✓	✓	✓
Individual Controls				✓	✓
Ideology Controls					✓
Adjusted R-squared	0.066	0.071	0.090	0.099	0.191
	Environmental policies support				
Panel (b): economic insecurity only	(1)	(2)	(3)	(4)	(5)
Economic insecurity	-0.088***	-0.081***	-0.055***	-0.052***	-0.052***
	(0.012)	(0.011)	(0.012)	(0.012)	(0.011)
Country-Year FE	✓	✓	✓	✓	✓
Regional Controls		✓	✓	✓	✓
Income Controls			✓	✓	✓
Individual Controls				✓	✓
Ideology Controls					✓
Adjusted R-squared	0.061	0.068	0.090	0.099	0.192
Regions	160	160	160	160	160
Observations	18,604	18,604	18,604	18,604	18,604

Notes: Columns (1)-(5) report the support for environmental policies regressed on income (Panel (a)) and perceived economic insecurity (Panel (b)). Controls include income or economic insecurity (1), aggregate regional controls (2), income controls (3), individual controls (4) and ideology controls (5). Country- and year-fixed effects are always included. Standard errors clustered at the regional level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

Table K.5: Regression: Support for environmental policies and perceived economic insecurity (tax vs green subsidies)

	OLS		Ordered probit	
	Pollution tax	Green subsidy	Pollution tax	Green subsidy
Variables	(1)	(2)	(3)	(4)
Economic insecurity	-0.003***	-0.001**	-0.002***	-0.001*
	(0.001)	(0.001)	(0.001)	(0.001)
Income	0.011**	0.001	0.010*	-0.000
	(0.006)	(0.005)	(0.005)	(0.005)
Climate change attitudes	0.015***	0.016***	0.015***	0.020***
	(0.001)	(0.001)	(0.001)	(0.001)
Cut 1			0.192	-1.323***
			(0.515)	(0.488)
Cut 2			1.014**	-0.738
			(0.514)	(0.486)
Cut 3			1.604***	-0.265
			(0.514)	(0.486)
Cut 4			2.726***	1.022**
			(0.518)	(0.486)
Regions	160	160	160	160
Observations	18,604	18,604	18,604	18,604
Adjusted R-squared	0.159	0.130	—	—

Notes: The table report the support for environmental taxes and green subsidy regressed on perceived economic insecurity. Columns (1)-(2) show the OLS specification. Columns (3)-(4) show the ordered probit specification. All controls and country- and year-fixed effects are always included. Standard errors clustered at the regional level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.

Table K.6: Regression: Support for redistributive policies and perceived economic insecurity

	Redistributive policies		Attitude towards redistribution	
	Basic income	EU social benefit scheme	Government should reduce income differences	Differences in standard of living should be small
Variables	(1)	(2)	(3)	(4)
Economic insecurity	0.003***	0.001**	0.004***	0.003***
	(0.000)	(0.001)	(0.001)	(0.001)
Income	-0.014***	-0.010**	-0.027***	-0.020***
	(0.004)	(0.004)	(0.005)	(0.006)
Regions	160	160	160	160
Observations	17,901	17,627	18,543	18,535
Adjusted R-squared	0.080	0.129	0.154	0.108

Notes: Column (1) includes a basic income question: "Overall, would you be against or in favour of having this scheme in [country]?". Answers from 1 ("Strongly against") to 4 ("Strongly in favour"). Column (2) includes the question "Overall, would you be against or in favour of having such a European Union-wide social benefit scheme?". Answers from 1 ("Strongly against") to 4 ("Strongly in favour"). Column (3) includes the question, "[...] to what extent you agree or disagree with each of the following statements. The government should take measures to reduce differences in income levels". Reversed answers from 1 ("Disagree strongly") to 4 ("Agree strongly"). Column (4) includes the question " [...] how much you agree or disagree with each of the following statements. For a society to be fair, differences in people's standard of living should be small". Reversed answers from 1 ("Disagree strongly") to 4 ("Agree strongly"). All controls and country- and year-fixed effects are always included. Standard errors clustered at the regional level in parentheses. ***, **, and * indicate levels of statistical significance at 1, 5, and 10 per cent, respectively.